In-Context Learning (and Unlearning) of Length Biases

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Abstract

Large language models have demonstrated strong capabilities to learn in-context, where exemplar input-output pairings are appended to the prompt for demonstration. However, existing work has demonstrated the ability of models to learn lexical and label biases in-context, which negatively impacts both performance and robustness of models. The impact of other statistical data biases remains underexplored, which this work aims to address. We specifically investigate the impact of length biases on in-context learning. We demonstrate that models do learn length biases in the context window for their predictions, and further empirically analyze the factors that modulate the level of bias exhibited by the model. In addition, we show that learning length information in-context can be used to counter the length bias that has been encoded in models (e.g., via fine-tuning). This reveals the power of in-context learning in debiasing model prediction behaviors without the need for costly parameter updates.

1 Introduction

In-context learning (ICL) has emerged as a new ability in large language models (LLMs), representative of a novel learning paradigm (Wei et al., 2022). With in-context learning, an LLM learns to perform an unseen task by seeing a number of demonstrations in the context window (Brown et al., 2020). Whereas previous methods such as fine-tuning update the model parameters to teach the model a desired task, ICL provides the model with input-output pairs as task exemplars directly at inference, with no parameter updates. While the goal of increased task accuracy is the same, the underlying mechanisms contributing to in-context learning are still being understood.

This has motivated a body of work aiming to understand how in-context learning works. Some works have demonstrated simlarities between finetuning and in-context learning. For example, instability due to the choice of examples occurs both in few-shot finetuning (Schick and Schütze, 2021; Gao et al., 2021) and in-context learning (Rubin et al., 2022; Liu et al., 2022; Wu et al., 2023). However, other work has shown counterintuitive results when comparing the apparent learning mechanisms of in-context learning and finetuning (Min et al., 2022).

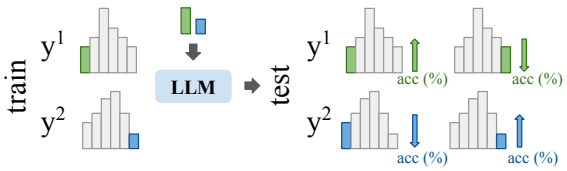


Figure 1: An illustration of our experiment setup and hypothesis. When sampling from the tails of the distribution (left of image), we introduce a data length bias. If the model can learn this shortcut feature in-context, we expect class performance on the data of similar length to be higher than data of the opposite length than what was seen in the context window (right of image).

A key area that is underexplored is whether in-context learning exhibits similar biases to finetuning with regard to statistical data biases. Statistical data biases can be defined as correlations between features and class labels. Under traditional learning paradigms such as fine-tuning, language models can learn exploitable statistical biases in the data. Such biases, or shallow features, can be exploited by a model as discriminatory features when they exhibit biased distributions across classes or are correlated with a specific class. This can lead to overestimates of a model's performance on the underlying task (Poliak et al., 2018; Gururangan et al., 2018).

Prior work has identified length as an exploitable statistical bias in natural inference

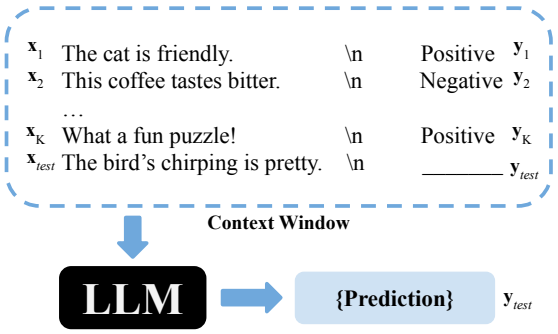


Figure 2: An overview of in-context learning using K input-output demonstrations concatenated to the test input $\{x_{test}, y_{test}\}$.

datasets. For example, in the MultiNLI and SNLI datasets, length has been shown to be a discriminatory feature (Gururangan et al., 2018), and on the ROC story cloze task choosing the longer ending performs above random baseline (Cai et al., 2017). However, length biases have been largely ignored in prior work on ICL, and some existing studies on which factors impact ICL have treated length as a static variable, selecting examples with similar lengths to test inputs (Min et al., 2022). It is therefore unclear whether models can exploit length biases in the data under an in-context learning setting (similar to finetuning) or whether length is another factor with counterintuitive tendencies.

In this work, we perform a series of empirical studies to investigate the ability of LLMs to learn statistical data biases in the context window during ICL (Figure 1). This has been studied in the finetuning literature, yet is underexplored in the ICL literature. We demonstrate empirically the ability of LLMs to learn length biases in-context. In the sections to follow, we analyze which factors influence these results, and we show the efficacy of ICL in debiasing finetuned models. Our results show that ICL can introduce biases to LLMs that negatively influence task performance. Specifically, our contributions are as follows:

1. We empirically demonstrate the ability of a range of LLM families to learn length biases in-context.

2. We investigate the influence of number of examples, number of model parameters, and class-length difference on how models learn biases.

3. We show that ICL can debias a model that contains existing length biases.

2 Background

In-context learning In-context learning is an emergent ability of LLMs that enables pre-trained models to learn an unseen task using a set of exemplars concatenated in the context window (see Figure 2). Formally, given a test example x, in-context learning concatenates K demonstration examples to the task instruction I, where $S = \{x_i, y_i\}_{i=1}^{K}$ denotes the example set. The performance of in-context learning, however, is highly dependent on both the selected examples (Rubin et al., 2022; Liu et al., 2022; Wu et al., 2023; Ye et al., 2023) and their orderings (Lu et al., 2022; Chen et al., 2023), resulting in performance variation from nearly random to comparable with finetuned models.

In-Context Learning & Bias While in-context learning has shown significant potential as a way to extract relevant information from an LLM and align the model with user expectations, it has also exhibited brittleness to an assortment of factors. These include selected examples (Rubin et al., 2022; Liu et al., 2022; Wu et al., 2023; Ye et al., 2023) and their orderings (Lu et al., 2022; Chen et al., 2023), which have recently been categorized under the umbrella of demonstration biases (Li et al., 2024).

Beyond demonstration bias, instability of ICL has been attributed to biases in the model toward predicting certain answers due to majority label bias, recency bias, and common token bias (Zhao et al., 2021). Correspondingly, several works have looked at identifying and mitigating label bias (Zhao et al., 2021; Fei et al., 2023) (Fei et al., 2023) with respect to lexical information, and Ali et al. (2024) have looked at mitigating "copy bias", where LLMs copy lexical information from demonstrations rather than learning underlying task information.

However, statistical data biases such as length information have been largely ignored in the in-context learning literature, yet received wide attention in the natural language inference literature with respect to traditional finetuned models (McCoy et al., 2019b; Poliak et al., 2018; Cai et al., 2017; Gururangan et al., 2018). Our work bridges this gap by looking at in-context learning with relation to a specific statistical bias: length bias.

3 Experiment Setup

In this section, we describe the experiment setup used in our analyses.

Datasets We use 7 binary classification datasets, representing natural language inference, sentiment analysis, and paraphrase detection tasks. As we sample from the tails of the length distributions, binary classification is ideal for our setting. For each dataset, we utilize the splits available from Huggingface. Dataset statistics are provided in Table 1, with detailed descriptions in subsection A.1. To count the length of each input, we use the NLTK word-tokenize package (Bird and Loper, 2004) rather than the LLM-specific tokenizers, to maintain consistency across experiments. Prompts are adapted from (Gao et al., 2023) and provided in subsection A.2.

Models Experiments in section 4 are run using five models from different LLM families, listed in Table 2. The selected models vary in size from 2.7B parameters to 8B parameters. Notably, the upper bound of the parameter range is due to our resource constraint, as each experiment is run using a single NVIDIA A100 GPU. For experiments in section 5 and section 6, we use a subset of these models, Llama3 and GPT-Neo. For experiments in subsection 5.1, we use the OPT model family.

Other Details Following Min et al. (2022), unless otherwise noted, all experiments use $k = 16$ demonstrations. For finetuning experiments, we use $k = 200$ finetuning examples. To minimize the impact of ordering effects, each result represents the mean of 4 trials, with standard deviation shown using error bars. Results are all run on the full validation split of each dataset.

In section 4, we investigate whether LLMs can learn length biases in-context. To further analyze these results, in section 5 we look at the impact of model parameter size, number of examples, and length distribution. Finally, section 6 demonstrates the utility of ICL to debias finetuned models that exhibit length biases.

4 Length Biases in Finetuning and ICL

In this section, we investigate the question ***do models learn length biases in-context?*** We demonstrate empirically the ability of LLMs to learn length biases in-context.

4.1 Method

Consider a dataset $D = \{(x_i, y_i)\}_{i=1}^{n}$ that contains n training instances. In this work, we consider binary classification datasets, where $Y = \{y^1, y^2\}$. We aim to introduce a distributional bias in the input lengths with respect to class. To introduce a length bias in k demonstrations drawn from D, we sample from the tails of each class's input length distribution. Specifically, we sample the top-$\frac{k}{2}$ examples belonging to y^1 and the bottom-$\frac{k}{2}$ from y^2 (and vice versa). This effectively produces a "worst-case scenario" in maximizing the distance between the classes under the given length distributions.

To provide a baseline for comparison, we compare against finetuning. Specifically, we finetune each model (using LoRA adapters (Hu et al., 2022)) on $k = 200$ training instances selected using the same procedure as above. As an additional baseline, for all experiments, we compare against randomly sampling the demonstrations and finetuning examples.

For results, we utilize a binning procedure. Specifically, we bin the validation set based on length, with $b = 6$ bins. In this setting, bin 1 represents the shortest 16.67% of validation instances and bin 6 represent the longest 16.67% validation instances across both classes. If a model has learned a length bias, for the validation class with the training set drawn from the shortest instances, we expect performance on bin 1 to be higher than bin 6, and vice versa for the validation class where the training set is drawn from the longest instances.

As a performance measurement, we report the sum of the individual accuracy from each class. As there may be a slight imbalance across classes in each bin, reporting individual class accuracy rather than the percentage of the entire bin ensures we account for class imbalances across bins.

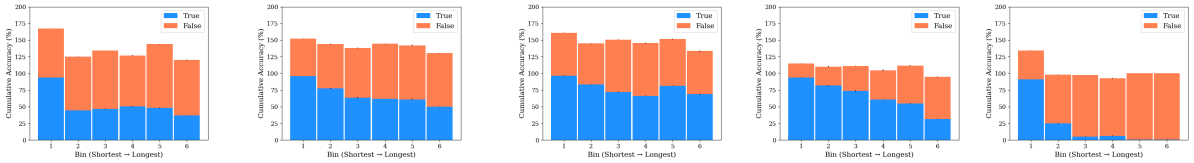
4.2 Results

We report results on HANS and PAWS-X_{EN} under finetuning (Figure 4 and Figure 6) and ICL (Figure 3 and Figure 5), where y_1 demonstrations were sampled from short instances and y_2 demonstrations were sampled from long instances. y_1 and y_2 correspond to the Blue and Orange bars, respectively. Our results show decreased performance

[1] (Dagan et al., 2006; Bar Haim et al., 2006; Giampiccolo et al., 2007; Bentivogli et al., 2009)

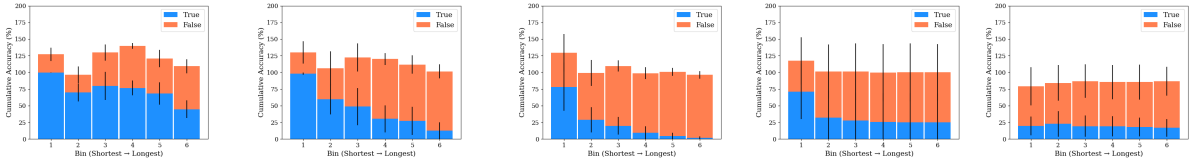
Category	Dataset	Task	#Train	#Val
Inference	QNLI (Wang et al., 2018)	Natural Language Inference	105k	5.46k
	RTE [1]	Natural Language Inference	2.49k	277
	WNLI (Levesque et al., 2011)	Natural Language Inference	635	71
	HANS (McCoy et al., 2019a)	Natural Language Inference	30k	30k
Single Sentence	SST-2 (Socher et al., 2013)	Sentiment Analysis	67.3k	872
Paraphrase Detection	MRPC (Dolan et al., 2004)	Paraphrase Detection	3.67k	408
	PAWS-X_{EN} (Yang et al., 2019)	Paraphrase Detection	49.4k	2k

Table 1: Datasets used in our experiments. We use the distributions available from Huggingface (Lhoest et al., 2021), and use the respective validation sets to measure performance. Dataset descriptions can be found in Table 3.



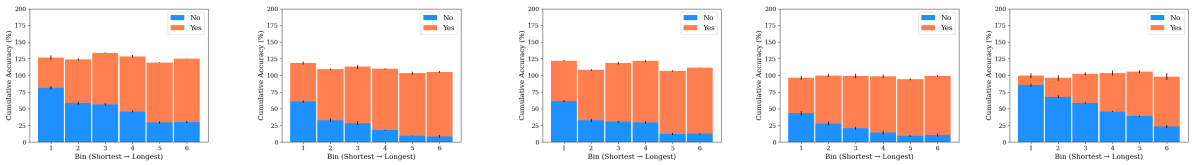
(a) Llama3 8B (b) Llama2 7B (c) Mistral 7B (d) OPT 6.7B (e) GPT Neo 2.7B

Figure 3: In-context learning validation performance across different models on the Hans dataset. For each graph, y_1 (Blue) was sampled from the short instances, and y_2 (Orange) was sampled from the long instances.



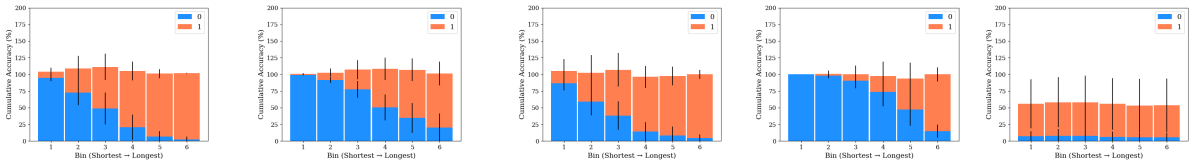
(a) Llama3 8B (b) Llama2 7B (c) Mistral 7B (d) OPT 6.7B (e) GPT Neo 2.7B

Figure 4: Finetuning validation performance across different models on the Hans dataset. For each graph, y_1 (Blue) was sampled from the short instances, and y_2 (Orange) was sampled from the long instances.



(a) Llama3 8B (b) Llama2 7B (c) Mistral 7B (d) OPT 6.7B (e) GPT Neo 2.7B

Figure 5: In-context learning validation performance across different models on the PAWS-X_{EN} dataset. For each graph, y_1 (Blue) was sampled from the short instances, and y_2 (Orange) was sampled from the long instances.



(a) Llama3 8B (b) Llama2 7B (c) Mistral 7B (d) OPT 6.7B (e) GPT Neo 2.7B

Figure 6: Finetuning validation performance across different models on the PAWS-X_{EN} dataset. For each graph, y_1 (Blue) was sampled from the short instances, and y_2 (Orange) was sampled from the long instances.

Model	Parameters
LLaMa 3 (Dubey et al., 2024)	8B
LLaMa 2 (Touvron et al., 2023)	7B
Mistral (Jiang et al., 2023)	7B
OPT (Zhang et al., 2022)	6.7B
GPT-Neo (Black et al., 2021)	2.7B

Table 2: Models used in section 4.

on validation examples that do not have a similar length as the demonstrations belonging to each respective class. This indicates that models can pick up length biases in-context. Additional results can be found in the Appendix.

5 Analysis of Influencing Factors

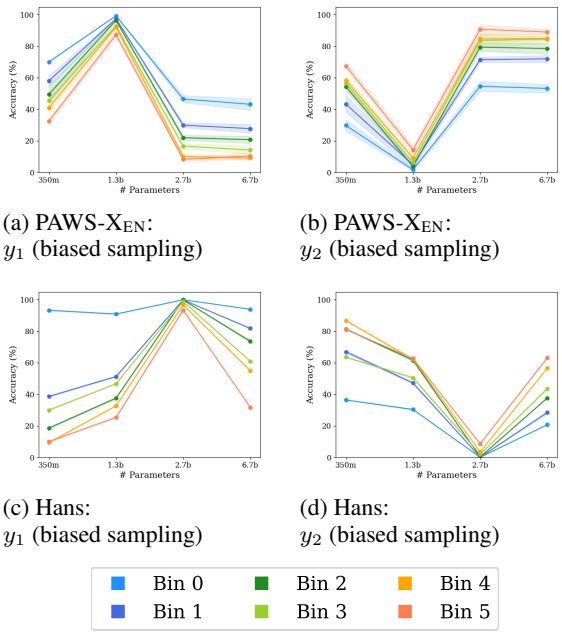
In this section, we investigate a further question of ***what factors influence how LLMs learn length biases in-context?*** We find that increased numbers of examples can exacerbate learned biases, and models across a range of sizes can learn length biases. Further, we find that length bias can be learned from as little as a few tokens of difference in average length between classes.

5.1 Number of Model Parameters

Existing work has suggested that the number of model parameters influences the ability of models to learn in-context, with larger models performing better (Milios et al., 2023; Lu et al., 2022). In this section, we investigate whether the number of parameters also influences the ability of models to learn length biases in-context. For example, if larger models are better at learning in-context, are they more susceptible or more resilient to learning statistical biases in the data?

We use the OPT model family (Zhang et al., 2022) across $p = \{350\text{M}, 1.3\text{B}, 2.7\text{B}, 6.7\text{B}\}$ parameters with $k = 16$ in-context examples. Note that the parameter count is upper-bounded based on computational resources. We use the procedure described in section 4 to introduce a length bias in the in-context demonstrations.

Results We report results on HANS and PAWS-X_{EN} in Figure 7. Notably, both datasets are designed to be challenging (see subsection A.1 for descriptions). Remaining datasets and conditions are reported in the Appendix. While we do observe length bias across varying model parameter sizes, there is not a consistent pattern of increased or decreased bias with increased model parameter sizes. Accordingly, we observe a dataset-model dependence with regard to the degree of length bias a model may learn.



(a) PAWS-X_{EN}: y_1 (biased sampling)

(b) PAWS-X_{EN}: y_2 (biased sampling)

(c) Hans: y_1 (biased sampling)

(d) Hans: y_2 (biased sampling)

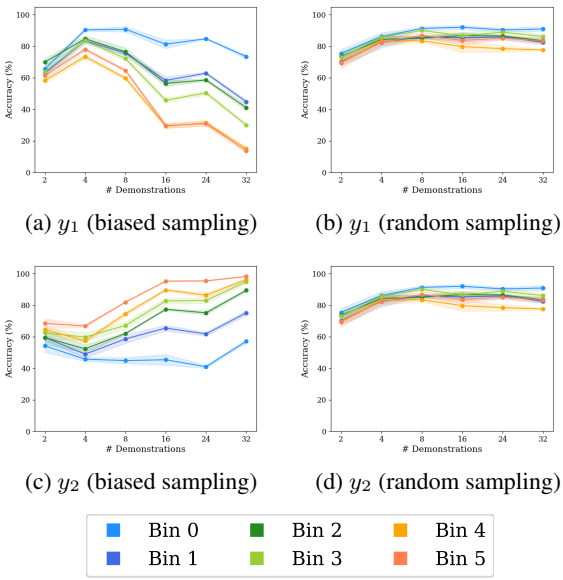
Figure 7: Validation performance across different numbers of model parameters using the OPT model family, on the PAWS-X_{EN} and Hans datasets. In this figure, in-context examples from y_2 were sampled from long instances, and in-context examples from y_1 were sampled from short instances. Each subfigure shows results on the validation instances in the respective class, with Bin 0 containing the shortest demonstrations and Bin 5 containing the longest demonstrations. Additional results can be found in the Appendix.

5.2 Number of Examples

The performance of ICL when using various numbers of examples has been studied in prior work (Wu et al., 2023; Min et al., 2022; Lu et al., 2022). As such, we investigate the sensitivity of LLM's ability to learn length bias across different numbers of in-context examples.

We use $k = \{2, 4, 8, 16, 24, 32\}$ in-context examples on the datasets in Table 1 using Llama3 8B and GPT Neo 2.7B. Following the procedure from section 4, we select the longest $\frac{k}{2}$ examples from y^1 and shortest $\frac{k}{2}$ examples from the y^2 (and vice versa), thereby introducing a bias in the length distribution of inputs across classes.

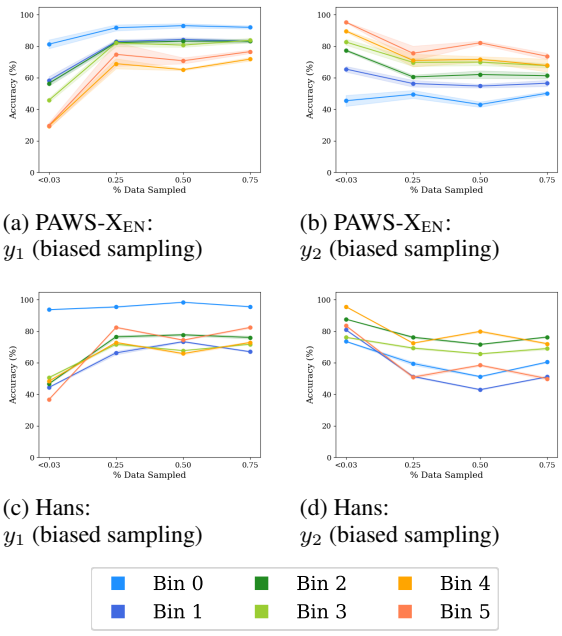
Results We report on the PAWS-X_{EN} dataset using Llama3 (8B) in Figure 8 and provide the average length for each class in subsection A.3. Our



(a) y_1 (biased sampling) (b) y_1 (random sampling)

(c) y_2 (biased sampling) (d) y_2 (random sampling)

Figure 8: Validation performance of Llama3 (8B) across different numbers of demonstrations on the PAWS-X_{EN} dataset. In this figure, in-context examples from y_2 were sampled from long instances, and in-context examples from y_1 were sampled from short instances. Each subfigure shows results on the validation instances in the respective class. The conditions that introduce length bias in the context window (a and c subfigures) demonstrate a larger spread between performance on short and long validation instances, indicating greater potential to learn bias with longer contexts. Additional results can be found in the Appendix.



(a) PAWS-X_{EN}: y_1 (biased sampling) (b) PAWS-X_{EN}: y_2 (biased sampling)

(c) Hans: y_1 (biased sampling) (d) Hans: y_2 (biased sampling)

Figure 9: Validation performance across different data sampling percentages using Llama 3 (8B), on the PAWS-X_{EN} and Hans datasets. In this figure, in-context examples from y_2 were sampled from long instances, and in-context examples from y_1 were sampled from short instances. Each subfigure shows results on the validation instances in the respective class, with Bin 0 containing the shortest demonstrations and Bin 5 containing the longest demonstrations. Additional results can be found in the Appendix.

results show that models can generally begin learning biases around 8 in-context examples, with the effect typically strengthening with increased numbers of examples.

Longer context models are gaining traction, with a recent line of work focusing on scaling in-context learning to larger numbers of demonstrations. Longer contexts can increase performance and decrease sensitivity to ordering effects (Cai et al., 2023; Hao et al., 2022), and contexts (beginning around $k = 8$) can decrease model calibration errors, where calibration is a measure of the faithfulness of a model's predictive uncertainty (Zhang et al., 2024). Our results demonstrate that longer contexts exhibit a greater potential for statistical data biases being learned in-context, and underscore the need for balanced selection methods with regard to potential data biases.

5.3 Difference in Average Demonstration Length Between Classes

Given the results from the previous section, we investigate whether the difference of the average demonstration length between classes influences the ability of LLMs to identify a length bias. We keep the number of examples consistent at $k = 16$ and sample from $p = \{0.25\%, 0.5\%, 0.75\%\}$ of the longest and shortest inputs for each class, respectively. For example, if y_1 is the long class, we sample $\frac{k}{2}$ instances from the longest $p = \{0.25\%, 0.5\%, 0.75\%\}$ of the instances belonging to y_1, and sample $\frac{k}{2}$ instances from the shortest $p = \{0.25\%, 0.5\%, 0.75\%\}$ of the instances belonging to y_2.

Results We report results using Llama 3 (8B) on the PAWS-X_{EN} dataset in Figure 9, where 0.03 corresponds to an approximate sampling percentage from the previous experiment setup. We observe a length bias across different sampling percentages, despite the decrease in difference between average class lengths (see Table 7). Intu-

itively, as the difference increases, so does the spread between performance across bins of different lengths. This indicates that while models can learn length biases from a few tokens difference (approximately 3 tokens on HANS under 0.75 sampling), the biases are amplified in the model as they are amplified in the demonstrations.

6 ICL for Debiasing Finetuned Models

In-context learning eliminates the need for expensive model parameter updates incurred when finetuning. However, it is often the case that a model may have encoded biases picked up from the pretraining and/or finetuning. As our previous experiments show that in-context learning can learn length information, a natural extension is to question whether ICL can be used to "unlearn" or mitigate previously learned length biases. In this section, we answer the question ***can ICL be used as an intervention to mitigate biases in finetuned models?***

We use previously finetuned models from section 4 and modify the length distribution to try to counteract the biases. Specifically we experiment with two conditions: 1) using in-context demonstrations drawn from the opposite tail of the length distribution from what was seen during finetuning, and 2) using randomly sampled in-context demonstrations. We again use Llama3 8B and GPT-Neo 2.7B for these experiments using the datasets in Table 1.

Results Results using Llama3 (8B) on HANS, PAWS-X_{EN}, and WNLI are reported in Figure 10, Figure 11 and Figure 12, respectively. We find that random sampling was able to counteract the bias, essentially "unlearning" the finetuned bias. This suggests that balanced data sampling is critical to minimize the likelihood of learning biases in-context. Further, if a dataset is balanced, random sampling may be sufficient. However, if a dataset contains shortcut features, more sophisticated sampling methods to mitigate the bias may be necessary.

Moreover, our results suggest balanced sampling over showing the models demonstrations of opposite lengths with respect to the finetuned bias. Specifically, the models learned the bias in the length information in the context window, regardless of how it contradicted what was seen during finetuning. One possible explanation is that the task may be implicitly encoded during pretraining and ICL extracts the knowledge (Xie et al., 2022; Min et al., 2022), however, further study is warranted on whether knowledge-extraction from ICL overrides knowledge-gain during finetuning.

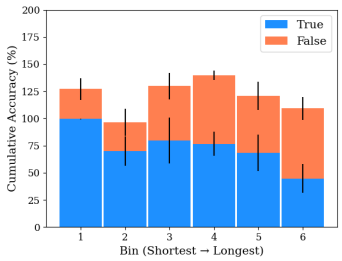
7 Discussion

In this work, we investigate the impact of demonstration length bias on model performance when learning in-context. Under this setting, length is a statistical data bias, where the shallow feature (length) is correlated with class labels. It is important to make the distinction, however, between length as a linguistic feature containing information relevant to the underlying task, and length as an artifact of the data collection protocol. For example, length is an informative syntactic feature in classifying truthfulness vs. deceptiveness (Yancheva and Rudzicz, 2013) and authorship detection (Yule, 1939), however, length biases can also arise from artifacts reflective of heuristics used by human data annotators (Gururangan et al., 2018). Our work pertains to the latter settings where length is an artifact rather than a task-informative natural language feature.

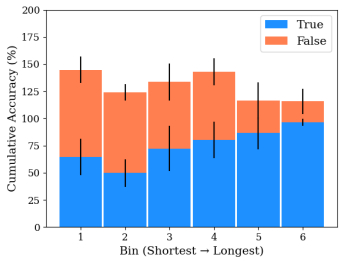
Which variables have the greatest impact on models learning length biases? In subsection A.3, we observe that when varying the number of in-context examples, the distance between classes is greater with fewer in-context examples. However, the amount of bias increases with increased numbers of examples. Further, while we observe that bias increases with increased length difference, we still observe learned bias when class length difference is reduced to as few as 3 tokens on the HANS dataset. This suggests that a key factor in learning bias is the number of examples the model sees. Additionally, our results suggest that any model can learn bias, and model parameter size is not necessarily correlated with increased ability to learn biases in-context.

8 Conclusion

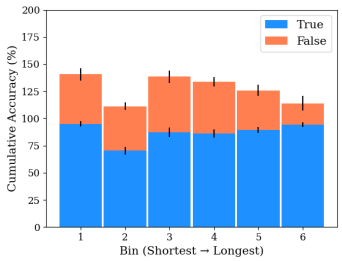
In this work, we empirically investigated the ability of LLMs to learn length biases under an in-context learning paradigm. Our results show that LLMs can learn statistical biases in the data. We further show the impact of model parameter sizes, number of examples, and class length difference on length biases. Finally, we demonstrate the potential for ICL to be used as a tool to debias fine-



(a) Finetuning: y_1 (Blue) short demonstrations, y_2 (Orange) long demonstrations.

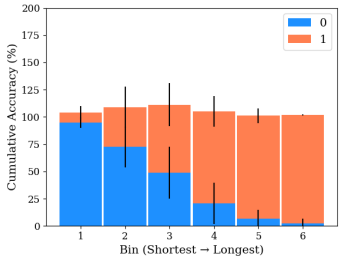


(b) Intervention: y_1 (Blue) long demonstrations, y_2 (Orange) short demonstrations.

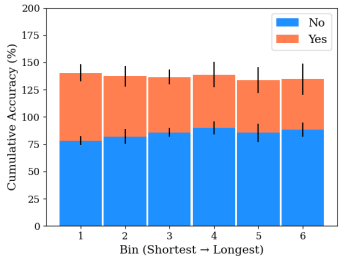


(c) Intervention: y_1 (Blue) and y_2 (Orange) demonstrations randomly sampled.

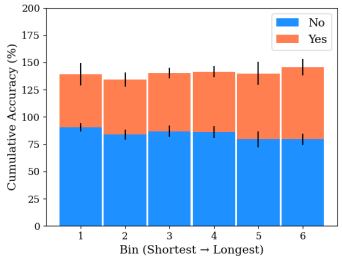
Figure 10: HANS validation set performance on a finetuned Llama 3 (8B) model exhibiting a length bias (see Figure 10a for finetuning performance prior to intervention). Figure 10b and Figure 10c (respectively) show results on two debiasing conditions: ICL demonstrations ($k = 16$) sampled from the opposite lengths from what the model saw during finetuning (i.e. y_1 long demonstrations, y_2 short demonstrations), and random sampling.



(a) Finetuning: y_1 (Blue) short demonstrations, y_2 (Orange) long demonstrations.

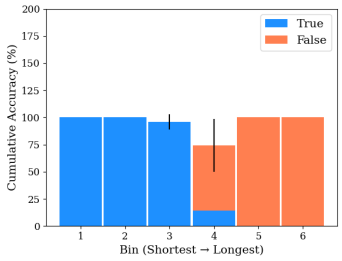


(b) Intervention: y_1 (Blue) long demonstrations, y_2 (Orange) short demonstrations.

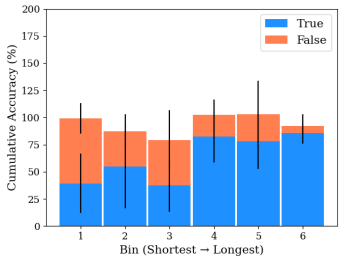


(c) Intervention: y_1 (Blue) and y_2 (Orange) demonstrations randomly sampled.

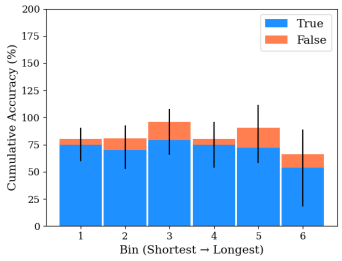
Figure 11: PAWS-X_{EN} validation set performance on a finetuned Llama 3 (8B) model exhibiting a length bias (see Figure 11a for finetuning performance prior to intervention). Figure 11b and Figure 11c (respectively) show results on two debiasing conditions: ICL demonstrations ($k = 16$) sampled from the opposite lengths from what the model saw during finetuning (i.e. y_1 long demonstrations, y_2 short demonstrations), and random sampling.



(a) Finetuning: y_1 (Blue) short demonstrations, y_2 (Orange) long demonstrations.



(b) Intervention: y_1 (Blue) long demonstrations, y_2 (Orange) short demonstrations.



(c) Intervention: y_1 (Blue) and y_2 (Orange) demonstrations randomly sampled.

Figure 12: WNLI validation set performance on a finetuned Llama 3 (8B) model exhibiting a length bias (see Figure 12a for finetuning performance prior to intervention). Figure 12b and Figure 12c (respectively) show results on two debiasing conditions: ICL demonstrations ($k = 16$) sampled from the opposite lengths from what the model saw during finetuning (i.e. y_1 long demonstrations, y_2 short demonstrations), and random sampling.

tuned models with previously learned length biases.

9 Limitations

While we test models up to 8B parameters, we acknowledge a limitation of this work is the pa-

rameter threshold due to available computational resources. We believe our results scale to larger models.

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A Appendix

A.1 Datasets

A.2 Prompts

A.3 Other Details

Binary NLI Datasets	
Dataset	**Description**
QNLI (Wang et al., 2018)	The Stanford Question Answering Dataset. A corpus of question-sentence pairs, with context sentences extracted from Wikipedia and questions written by a human annotator. The model is tasked with determining whether the context contains the answer to (entails) the question.
RTE [2]	The Recognizing Textual Entailment (RTE) datasets. A corpus constructed from annual textual entailment challenges based on news and Wikipedia text.
WNLI (Levesque et al., 2011)	The Winograd Schema Challenge. A corpus of reading comprehension sentence pairs, where ambiguous pronouns are replaced with each possible referent. The task is to predict if the substituted sentence is entailed by the original sentence.
HANS (McCoy et al., 2019a)	Heuristic Analysis for NLI Systems. A corpus of challenging premise and hypothesis pairs designed to target evaluation of lexical overlap, sub-sequence, and constituent heuristics.
Single-Sentence Datasets	
Dataset	**Description**
SST-2 (Socher et al., 2013)	The Stanford Sentiment Treebank. A corpus of sentences extracted from movie reviews, with human judgments of positive or negative sentiment.
Similarity & Paraphrase Detection Datasets	
Dataset	**Description**
MRPC (Dolan et al., 2004)	The Microsoft Research Paraphrase Corpus. A corpus of sentence pairs extracted from online news sources. Human raters judged semantic equivalence.
PAWS-X_{EN} (Yang et al., 2019)	Paraphrase Adversaries from Word Scrambling - Cross-lingual. A corpus of challenging paraphrase and non-paraphrase pairs created using data from Wikipedia. Sentence pairs were generated using controlled word swapping and back translation to ensure high lexical overlap, and human raters judged semantic equivalence. Our experiments utilize the English data split.

Table 3: Dataset descriptions.

Binary NLI Datasets	
Dataset	**Prompt**
QNLI (Wang et al., 2018)	{SENTENCE} QUESTION: {QUESTION} TRUE OR FALSE? ANSWER:
RTE [3]	{SENTENCE1} QUESTION: {SENTENCE2} TRUE OR FALSE? ANSWER:
WNLI (Levesque et al., 2011)	{SENTENCE1} QUESTION: {SENTENCE2} TRUE OR FALSE? ANSWER:
HANS (McCoy et al., 2019a)	{PREMISE} QUESTION: {HYPOTHESIS} TRUE OR FALSE? ANSWER:
Single-Sentence Datasets	
Dataset	**Description**
SST-2 (Socher et al., 2013)	{SENTENCE} QUESTION: IS THIS SENTENCE POSITIVE OR NEGATIVE? ANSWER:
Similarity & Paraphrase Detection Datasets	
Dataset	**Description**
MRPC (Dolan et al., 2004)	SENTENCE 1: {SENTENCE1} SENTENCE 2: {SENTENCE2} QUESTION: DO BOTH SENTENCES MEAN THE SAME THING? ANSWER:
PAWS-X_{EN} (Yang et al., 2019)	SENTENCE 1: {SENTENCE1} SENTENCE 2: {SENTENCE2} QUESTION: DO BOTH SENTENCES MEAN THE SAME THING? ANSWER:

Table 4: Prompts used in our experiments.

Dataset	Condition	Class	# Examples					
			2	4	8	16	24	32
QNLI	Random	True	51.33	42.5.00	47.67	44.81	48.48	45.49
		False	48.60	36.5	49.94	47.70	46.86	48.44
	False-L	True	14.00	14.00	14.75	15.38	15.75	16.06
		False	227.00	207.50	191.50	175.00	165.75	159.50
	True-L	True	446.00	445.40	445.00	358.63	305.92	276.00
		False	15.00	15.50	15.75	15.88	16.00	16.25
RTE	Random	True	60.00	59.33	61.25	66.57	63.02	65.77
		False	44.00	57.71	53.00	63.06	56.10	63.36
	False-L	True	17.00	17.50	18.5	19.63	20.17	20.75
		False	277.00	253.00	233.75	216.75	206.83	200.75
	True-L	True	195.00	194.00	192.00	187.25	183.67	181.19
		False	16.00	17.50	18.75	19.75	20.75	21.50
WNLI	Random	True	37.75	45.50	41.07	41.33	38.20	39.62
		False	41.75	38.17	39.50	35.18	36.65	36.72
	False-L	True	19.00	19.50	19.75	20.38	20.83	21.25
		False	92.00	91.50	87.75	82.38	79.67	77.13
	True-L	True	93.00	91.50	89.50	84.88	82.42	80.63
		False	19.00	19.00	19.50	20.13	20.58	21.00
HANS	Random	True	18.80	19.25	21.50	20.21	20.17	20.91
		False	20.67	21.38	21.22	21.31	21.41	21.05
	False-L	True	13.00	13.00	13.00	13.00	13.00	13.00
		False	27.00	27.00	27.00	27.00	27.00	27.00
	True-L	True	27.00	27.00	27.00	27.00	27.00	27.00
		False	15.00	15.00	15.00	15.00	15.00	15.00
SST-2	Random	True	21.00	21.00	20.85	19.21	19.62	19.30
		False	22.33	18.00	19.16	18.00	17.78	19.16
	False-L	True	10.00	10.00	10.00	10.00	10.00	10.00
		False	61.00	61.00	60.50	60.00	59.58	59.19
	True-L	True	62.00	61.50	61.00	60.00	59.08	58.44
		False	10.00	10.00	10.00	10.00	10.00	10.00
MRPC	Random	True	59.00	57.00	59.33	58.67	59.13	60.22
		False	57.29	59.91	58.61	63.06	62.06	60.46
	False-L	True	34.00	34.00	34.50	35.13	35.67	36.19
		False	97.00	91.50	88.25	86.00	85.00	84.25
	True-L	True	82.00	81.50	81.25	80.25	79.50	79.00
		False	32.00	32.00	33.00	33.50	34.00	34.69
PAWS-X_{EN}	Random	True	63.33	57.50	58.50	57.84	58.11	58.00
		False	61.60	58.60	58.06	54.31	58.79	59.42
	False-L	True	26.00	27.00	27.50	28.25	28.50	28.63
		False	85.00	85.00	84.50	83.50	83.00	82.50
	True-L	True	86.00	86.00	86.00	85.63	85.25	84.94
		False	26.00	26.00	26.25	27.63	28.33	28.75

Table 5: Average input length (including prompt length) across different numbers of examples.

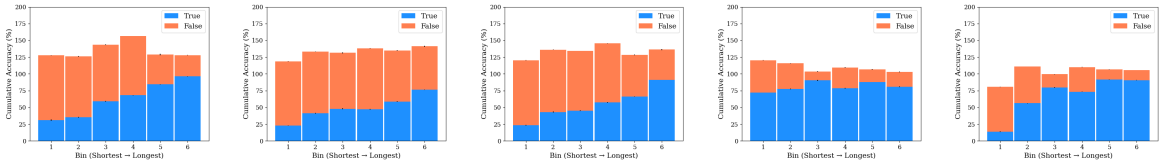
Dataset	Class	Average Length
QNLI	True	50.03
	False	47.50
RTE	True	63.25
	False	64.99
WNLI	True	40.23
	False	37.00
HANS	True	20.53
	False	20.99
SST2	True	28.17
	False	28.93
MRPC	True	57.72
	False	61.16
PAWS-X_{EN}	True	58.63
	False	58.63

Table 6: Average input length for each validation set. Reported input lengths include the prompt length (consistent across all inputs); prompts can be found in subsection A.2.

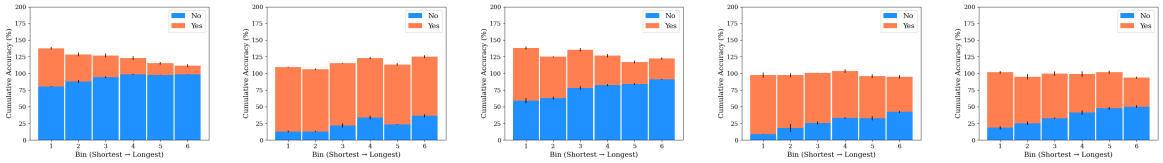
Dataset	Condition	Class	% Sampled		
			0.25	0.50	0.75
QNLI	False-L	True	31.35	36.27	40.91
		False	65.87	57.55	52.05
	True-L	True	69.98	60.16	53.93
		False	31.56	36.33	40.61
RTE	False-L	True	33.05	40.03	46.23
		False	122.98	90.66	76.35
	True-L	True	124.51	91.02	77.78
		False	33.44	39.98	47.21
WNLI	False-L	True	25.23	28.01	31.70
		False	56.01	47.21	40.60
	True-L	True	60.08	48.73	43.12
		False	24.72	28.12	31.05
HANS	False-L	True	16.92	18.72	19.65
		False	23.59	22.65	21.97
	True-L	True	23.29	22.39	21.77
		False	18.17	19.40	20.16
SST2	False-L	True	11.07	12.66	14.92
		False	29.57	23.90	20.36
	True-L	True	30.55	25.14	21.37
		False	11.06	12.26	14.27
MRPC	False-L	True	44.89	49.24	53.12
		False	74.31	69.98	65.67
	True-L	True	70.53	65.61	61.50
		False	47.12	52.35	56.78
PAWS-X_{EN}	False-L	True	44.61	49.51	53.98
		False	72.44	67.80	63.31
	True-L	True	72.58	67.93	63.46
		False	44.56	49.53	54.04

Table 7: Average input length across different sampling bins (by percentage of data sampled from). Reported input lengths include the prompt length (consistent across all inputs); prompts can be found in subsection A.2.

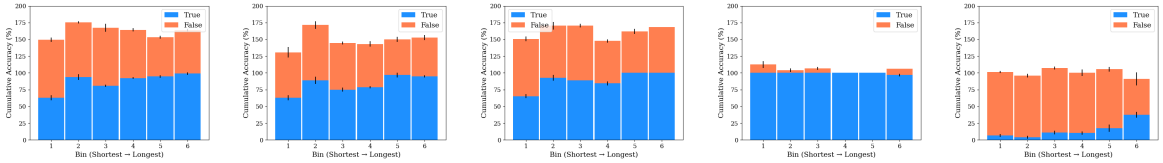
A.4 Additional Length Bias Results



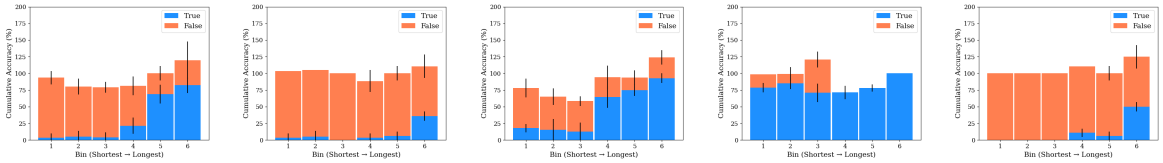
(a) Hans



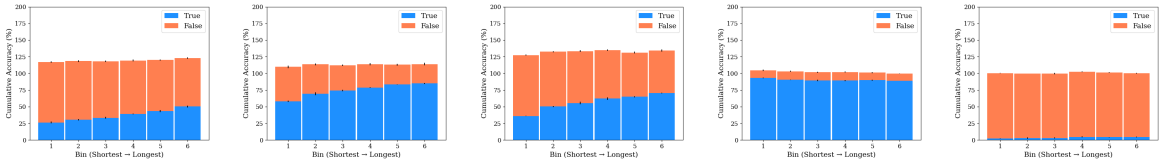
(b) PAWS-X_{EN}



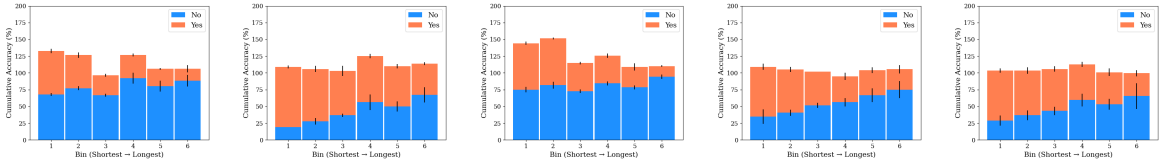
(c) RTE



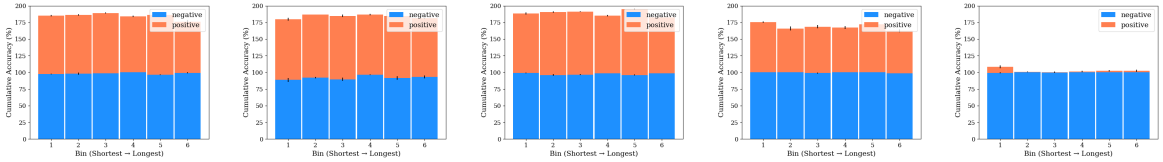
(d) WNLI



(e) QNLI

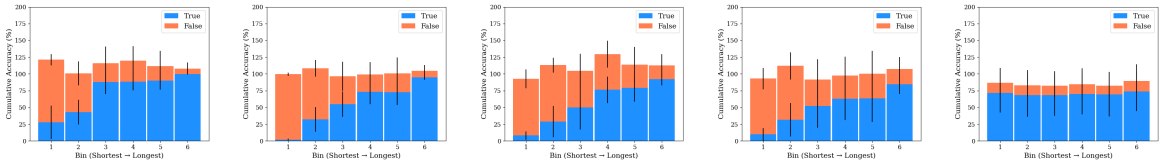


(f) MRPC

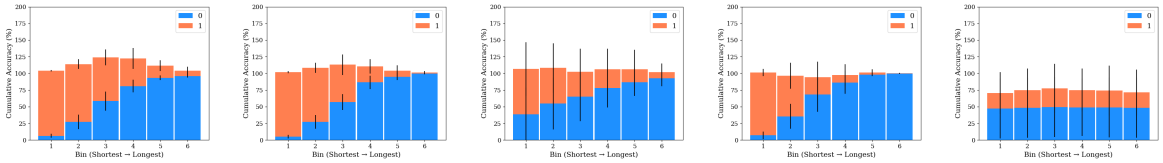


(g) SST-2

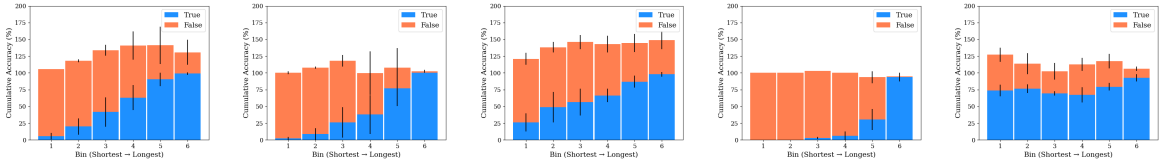
Figure 13: ICL performance of Llama3 8B, Llama2 7B, Mistral 7B, OPT 6.7B, and GPT Neo 2.7B (from left to right) where y_1 (Blue) samples long demonstrations and y_2 (Orange) samples short demonstrations.



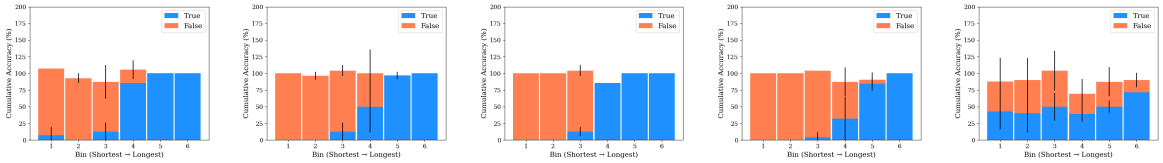
(a) Hans



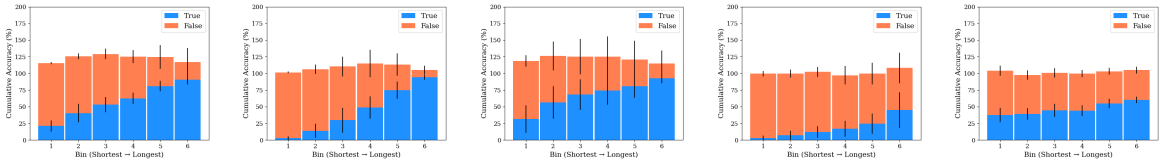
(b) PAWS-X_{EN}



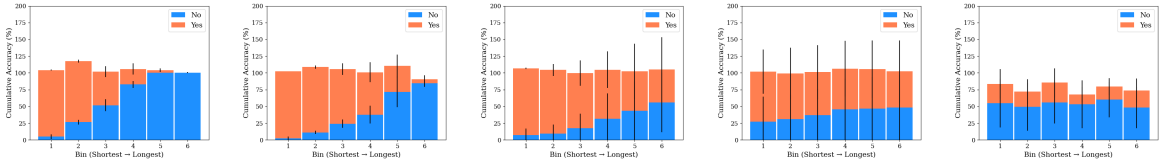
(c) RTE



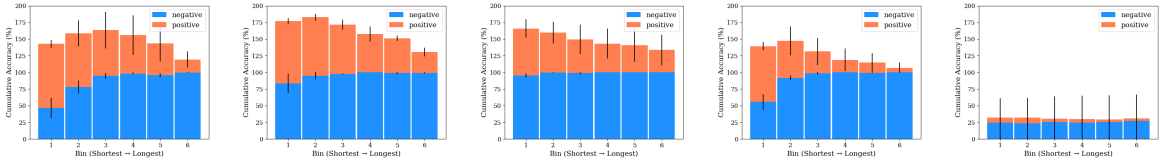
(d) WNLI



(e) QNLI

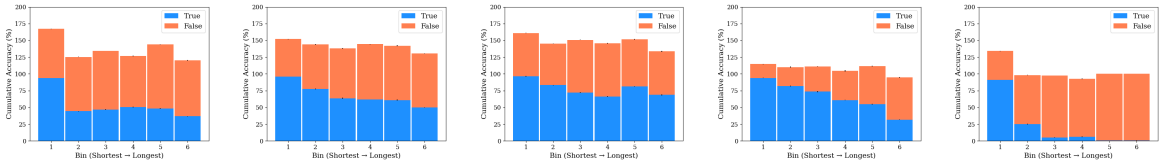


(f) MRPC

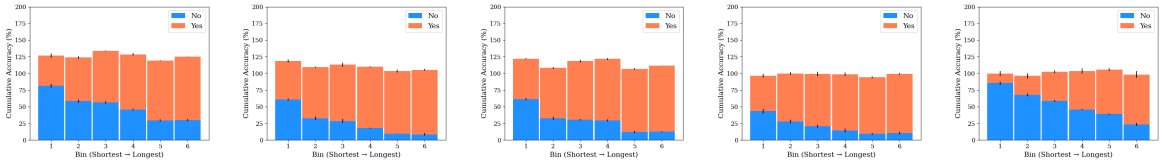


(g) SST-2

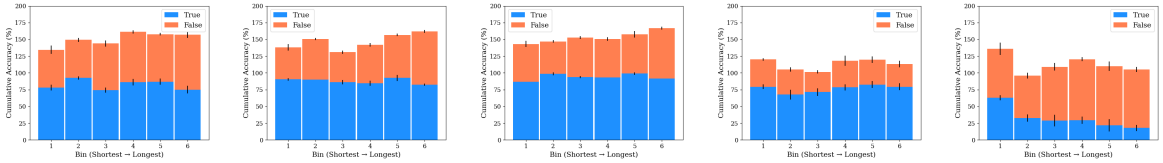
Figure 14: Finetuning performance of Llama3 8B, Llama2 7B, Mistral 7B, OPT 6.7B, and GPT Neo 2.7B (from left to right) where y_1 (Blue) samples long demonstrations and y_2 (Orange) samples short demonstrations.



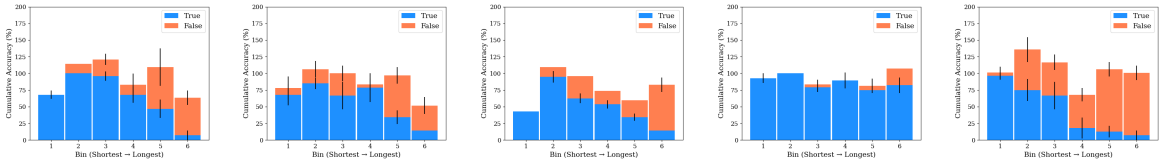
(a) Hans



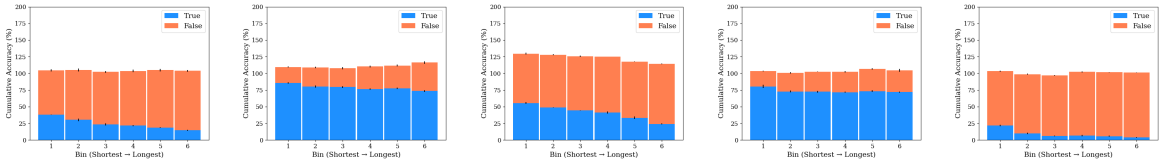
(b) PAWS-X_{EN}



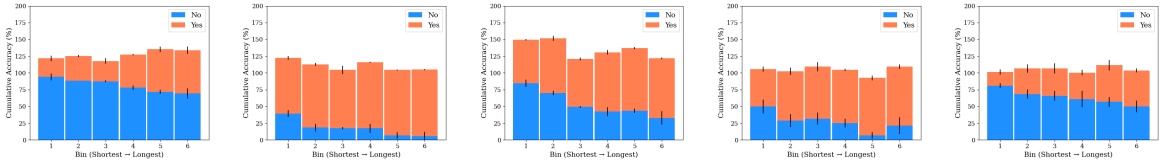
(c) RTE



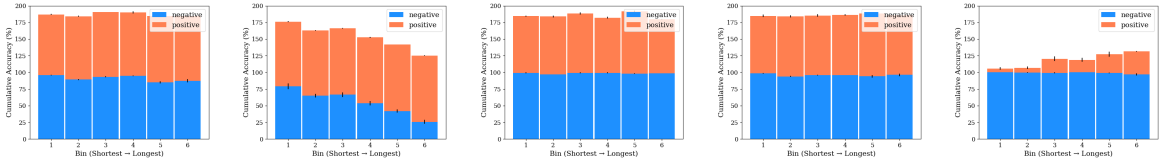
(d) WNLI



(e) QNLI

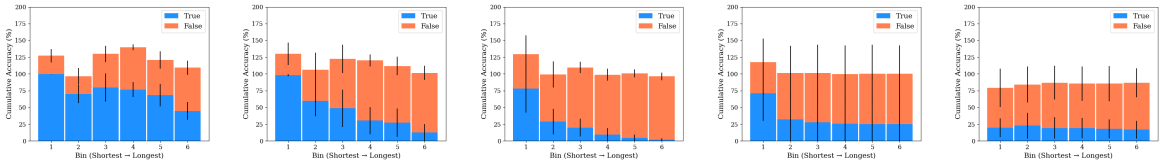


(f) MRPC

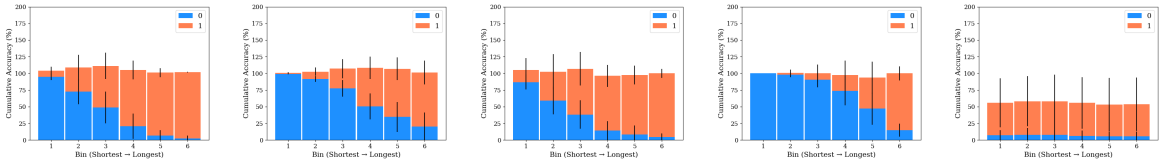


(g) SST-2

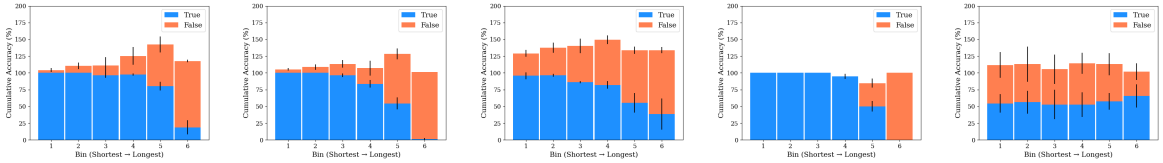
Figure 15: ICL performance of Llama3 8B, Llama2 7B, Mistral 7B, OPT 6.7B, and GPT Neo 2.7B (from left to right) where y_1 (Blue) samples short demonstrations and y_2 (Orange) samples long demonstrations.



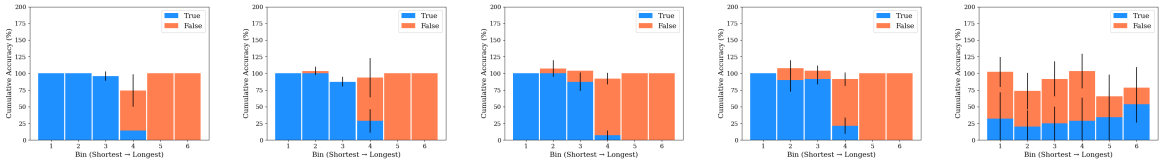
(a) Hans



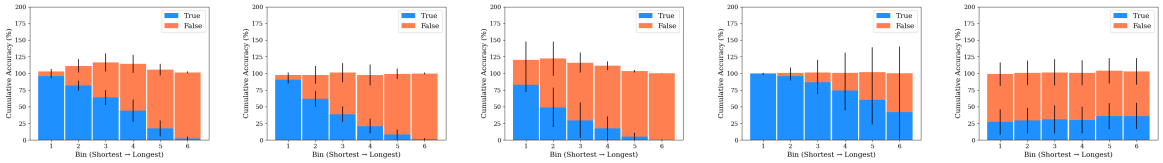
(b) PAWS-X_{EN}



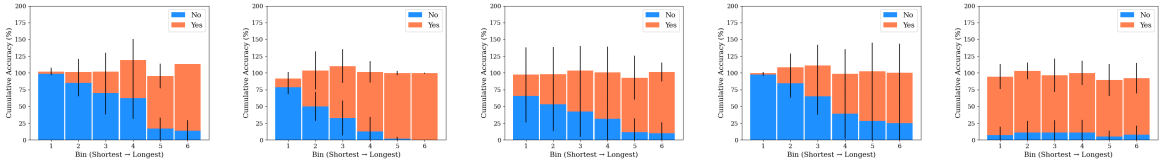
(c) RTE



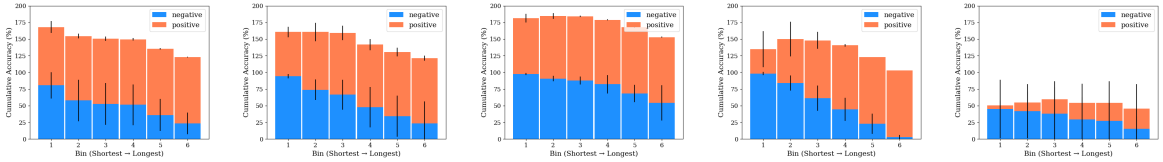
(d) WNLI



(e) QNLI

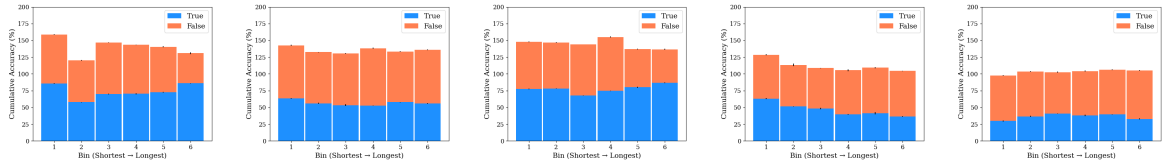


(f) MRPC

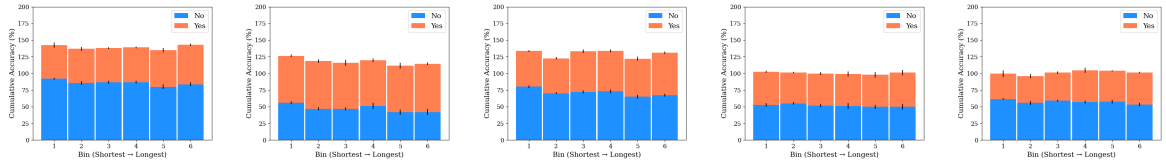


(g) SST-2

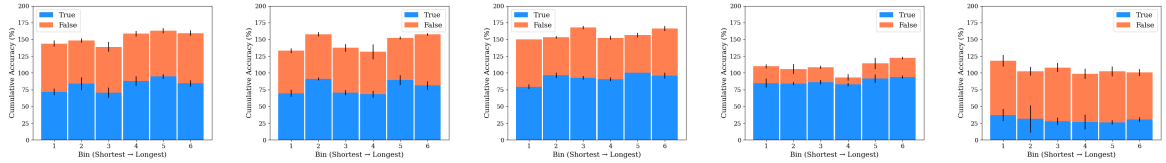
Figure 16: Finetuning performance of Llama3 8B, Llama2 7B, Mistral 7B, OPT 6.7B, and GPT Neo 2.7B (from left to right) where y_1 (Blue) samples short demonstrations and y_2 (Orange) samples long demonstrations.



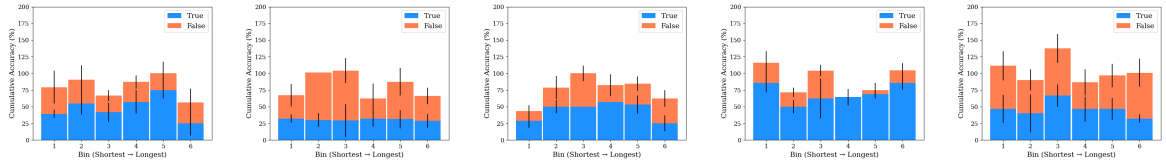
(a) Hans



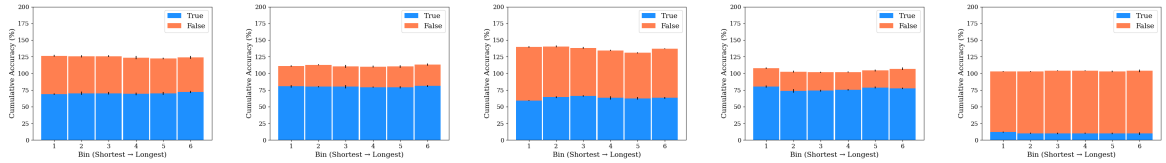
(b) PAWS-X_{EN}



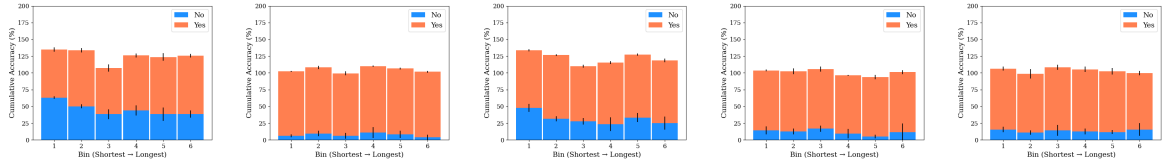
(c) RTE



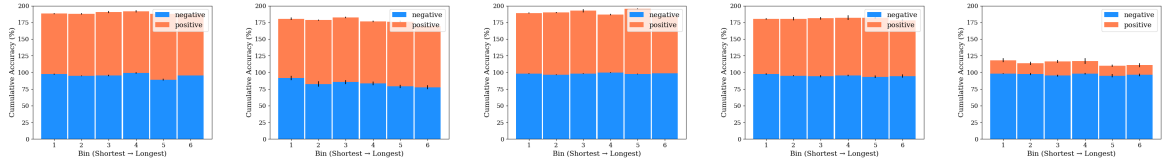
(d) WNLI



(e) QNLI

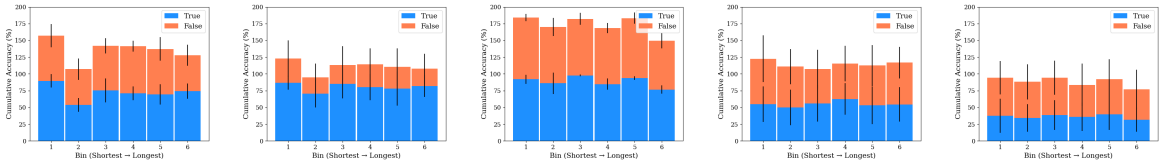


(f) MRPC

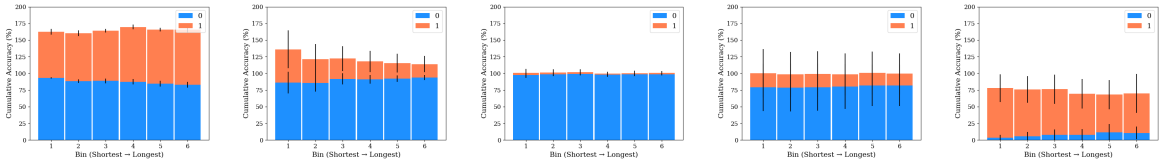


(g) SST-2

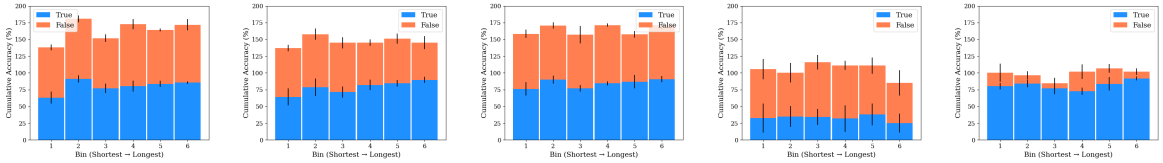
Figure 17: ICL performance of Llama3 8B, Llama2 7B, Mistral 7B, OPT 6.7B, and GPT Neo 2.7B (from left to right) where y_1 (Blue) and y_2 (Orange) are both randomly sampled.



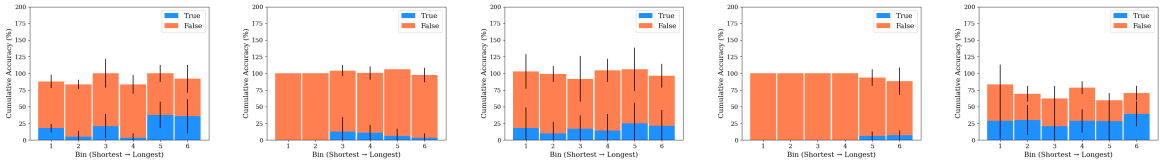
(a) Hans



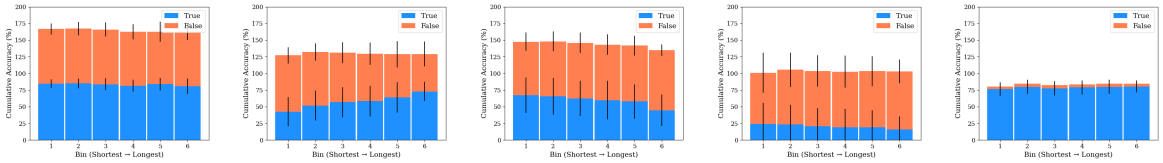
(b) PAWS-X_{EN}



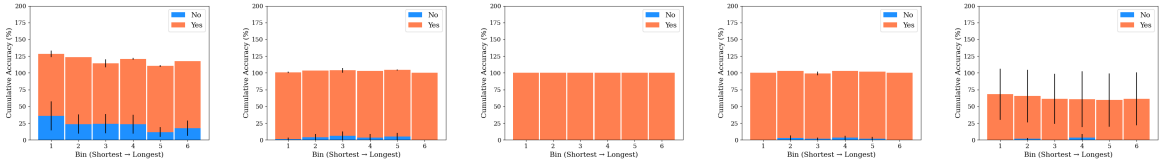
(c) RTE



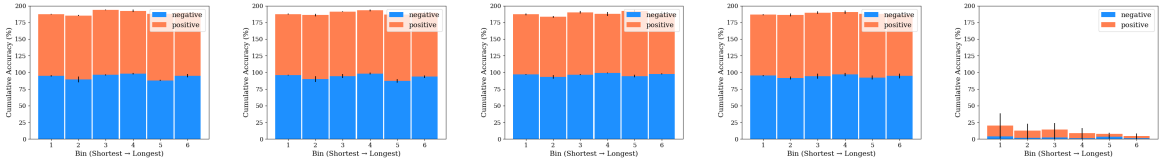
(d) WNLI



(e) QNLI



(f) MRPC

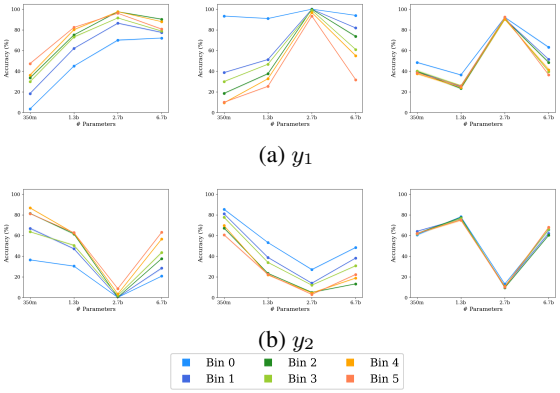


(g) SST-2

Figure 18: Finetuning performance of Llama3 8B, Llama2 7B, Mistral 7B, OPT 6.7B, and GPT Neo 2.7B (from left to right) where y_1 (Blue) and y_2 (Orange) are both randomly sampled.

A.5 Additional Model Parameter (OPT) Results

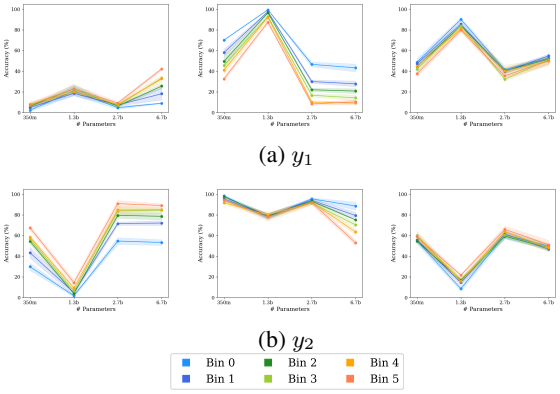
Each of the following figures shows validation performance when varying the number of model parameters using the OPT model family. Bin 0 contains the shortest demonstrations and Bin 5 contains the longest demonstrations. Each subfigure shows the validation accuracy on a single class when in-context instances belonging to the respective class were sampled from long instances, short instances, and randomly sampled (left to right).



(a) y_1

(b) y_2

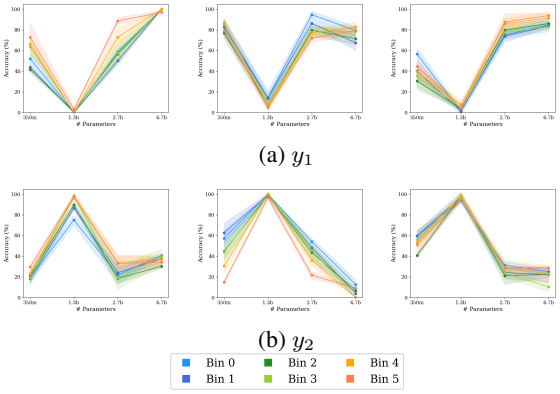
Figure 19: Hans dataset (OPT)



(a) y_1

(b) y_2

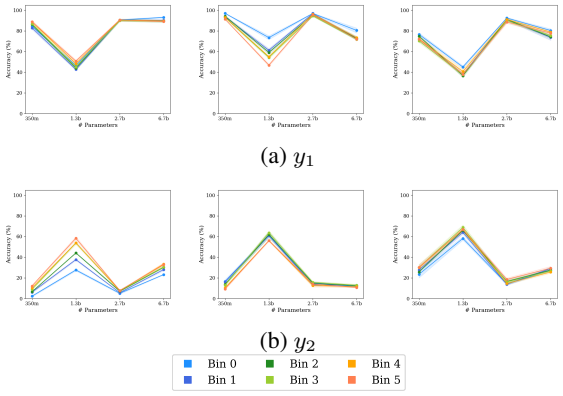
Figure 20: PAWS-X_{EN} dataset (OPT)



(a) y_1

(b) y_2

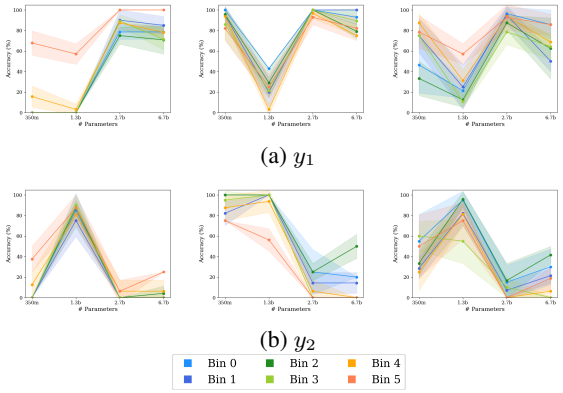
Figure 21: RTE dataset (OPT)



(a) y_1

(b) y_2

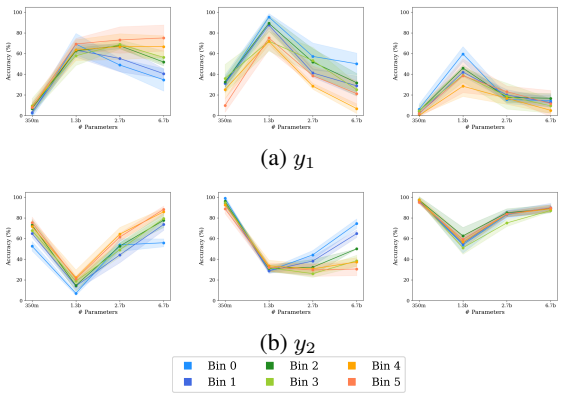
Figure 22: QNLI dataset (OPT)



(a) y_1

(b) y_2

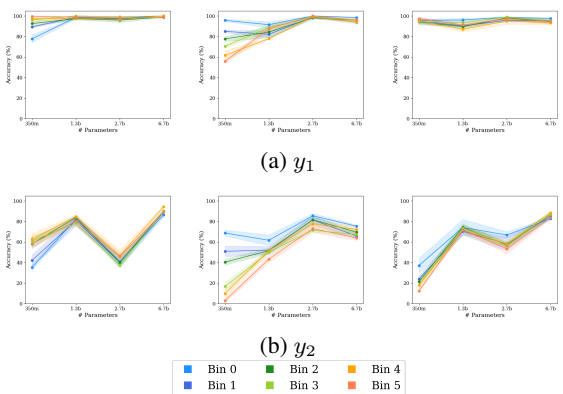
Figure 23: WNLI dataset (OPT)



(a) y_1

(b) y_2

Figure 24: MPRC dataset (OPT)



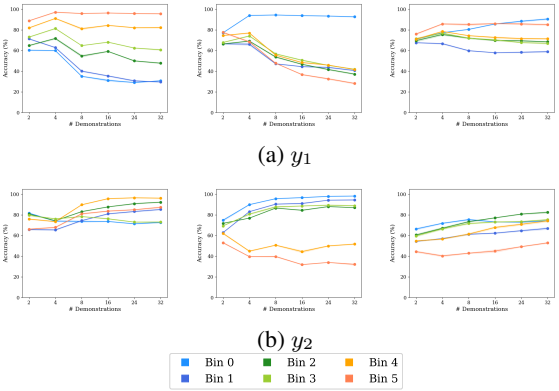
(a) y_1

(b) y_2

Figure 25: SST-2 dataset (OPT)

A.6 Additional Number of Examples Results

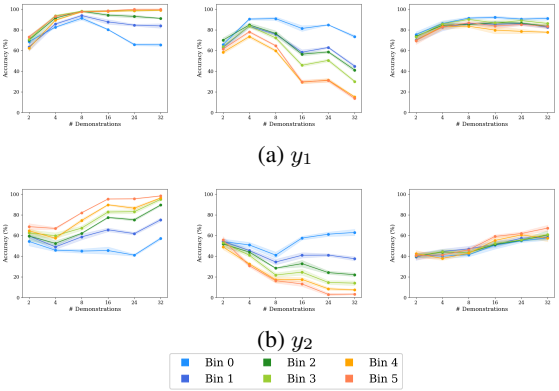
Each of the following figures shows validation performance when varying the number of examples using Llama3 8B and GPT Neo 2.7B. Bin 0 contains the shortest demonstrations and Bin 5 contains the longest demonstrations. Each subfigure shows the validation accuracy on a single class when in-context instances belonging to the respective class were sampled from long instances, short instances, and randomly sampled (left to right).



(a) y_1

(b) y_2

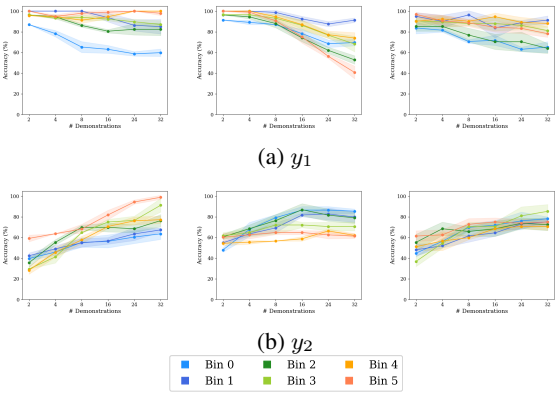
Figure 26: Hans dataset (Llama 3 8B)



(a) y_1

(b) y_2

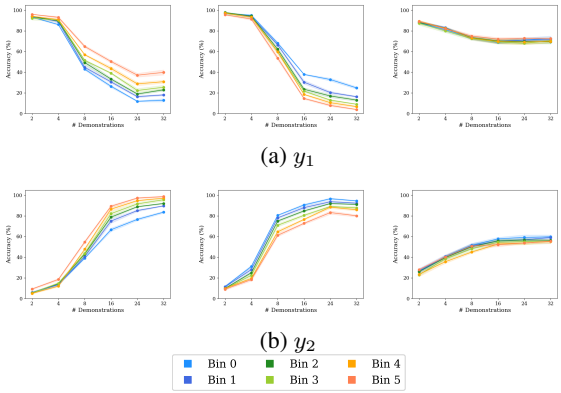
Figure 27: PAWS-X_{EN} dataset (Llama 3 8B)



(a) y_1

(b) y_2

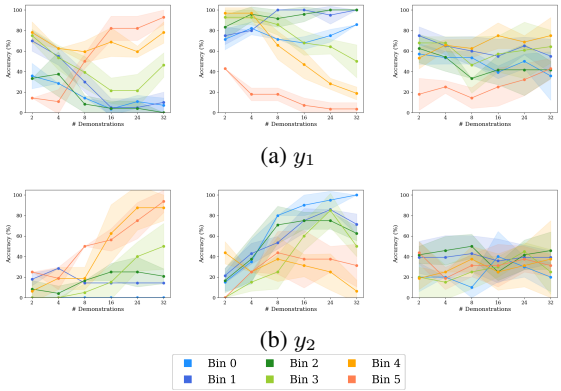
Figure 28: RTE dataset (Llama 3 8B)



(a) y_1

(b) y_2

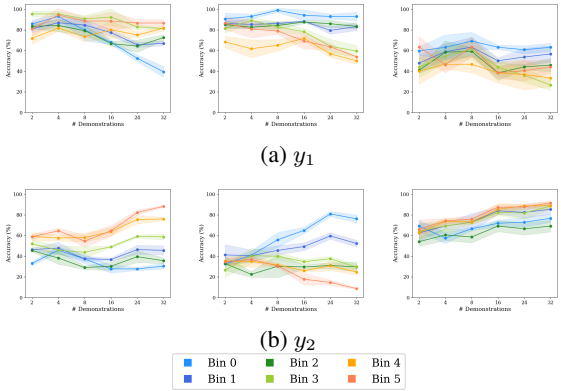
Figure 29: QNLI dataset (Llama 3 8B)



(a) y_1

(b) y_2

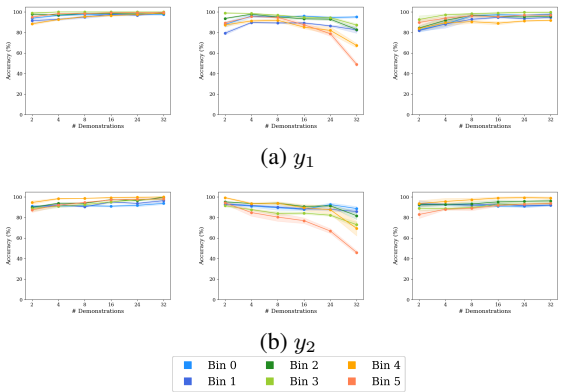
Figure 30: WNLI dataset (Llama 3 8B)



(a) y_1

(b) y_2

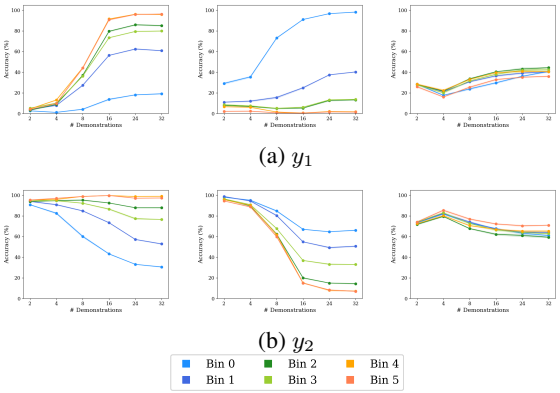
Figure 31: MPRC dataset (Llama 3 8B)



(a) y_1

(b) y_2

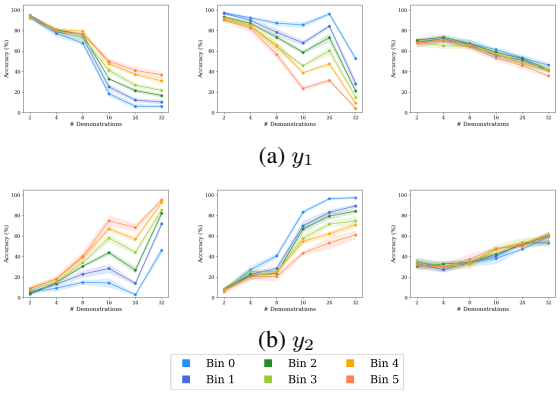
Figure 32: SST-2 dataset (Llama 3 8B)



(a) y_1

(b) y_2

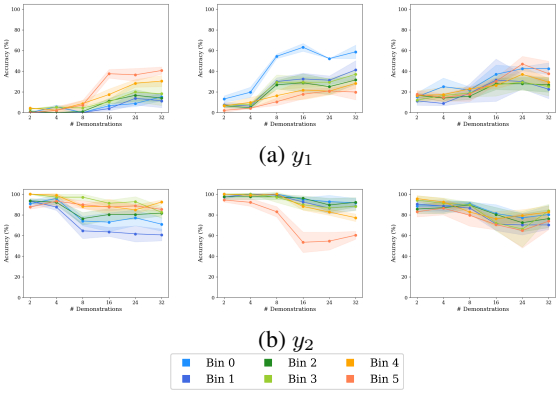
Figure 33: Hans dataset (GPT Neo 2.7B)



(a) y_1

(b) y_2

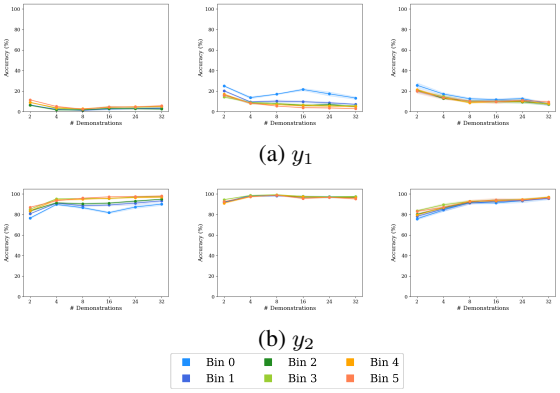
Figure 34: PAWS-X_{EN} dataset (GPT Neo 2.7B)



(a) y_1

(b) y_2

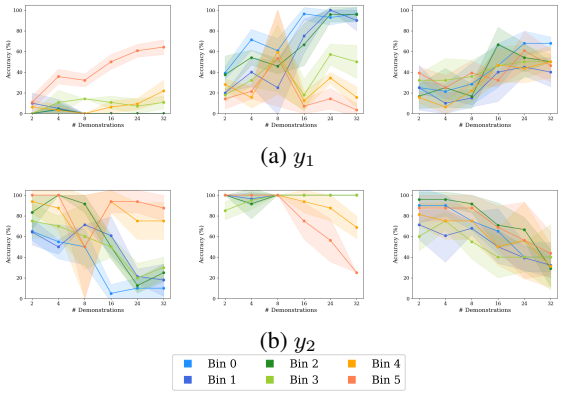
Figure 35: RTE dataset (GPT Neo 2.7B)



(a) y_1

(b) y_2

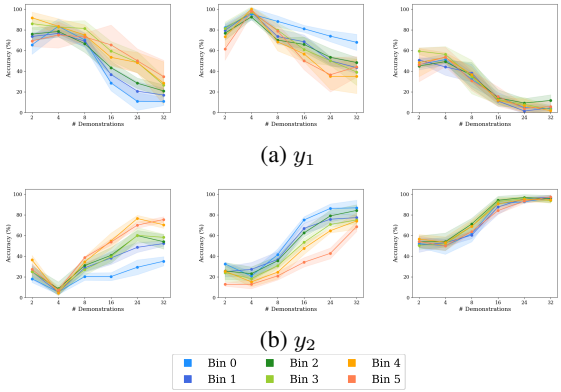
Figure 36: QNLI dataset (GPT Neo 2.7B)



(a) y_1

(b) y_2

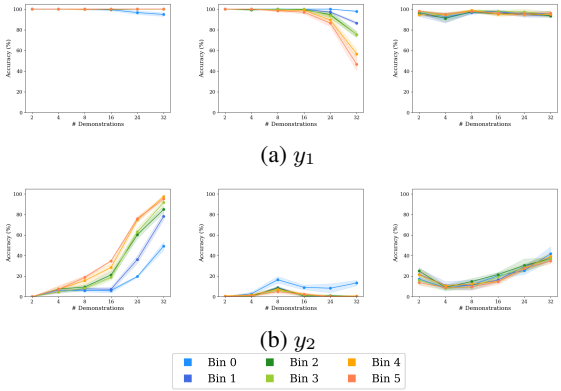
Figure 37: WNLI dataset (GPT Neo 2.7B)



(a) y_1

(b) y_2

Figure 38: MPRC dataset (GPT Neo 2.7B)



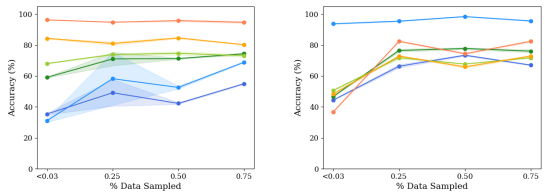
(a) y_1

(b) y_2

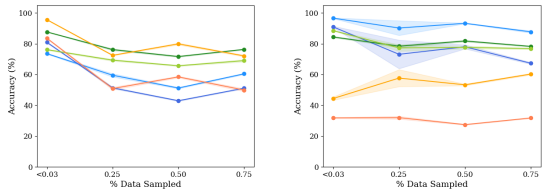
Figure 39: SST-2 dataset (GPT Neo 2.7B)

A.7 Additional Length Difference Results

Each of the following figures shows validation performance when varying the sampling percentage from each class using Llama3 8B and GPT Neo 2.7B. Bin 0 contains the shortest demonstrations and Bin 5 contains the longest demonstrations. Each subfigure shows the validation accuracy on a single class when in-context instances belonging to the respective class were sampled from long instances (left) and short instances (right).

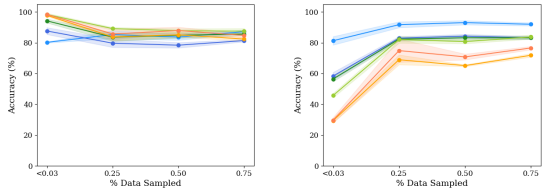


(a) y_1

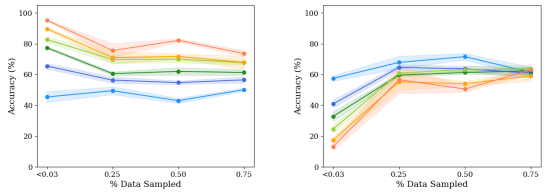


(b) y_2

Figure 40: Hans dataset (Llama 3 8B)

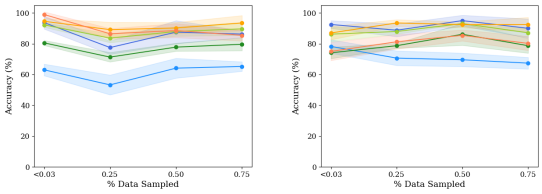


(a) y_1

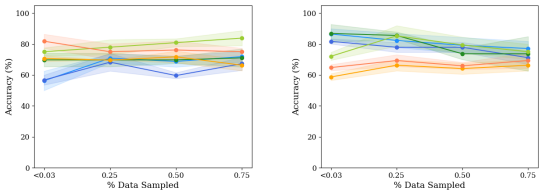


(b) y_2

Figure 41: PAWS-X_{EN} dataset (Llama 3 8B)

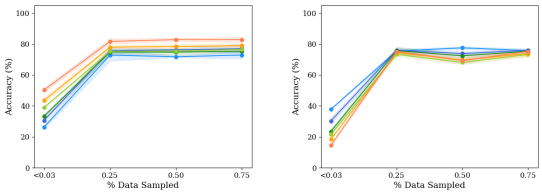


(a) y_1

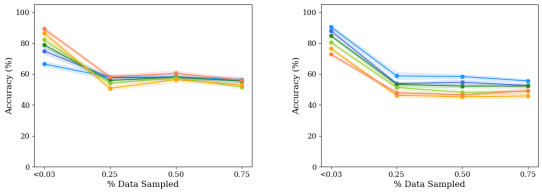


(b) y_2

Figure 42: RTE dataset (Llama 3 8B)

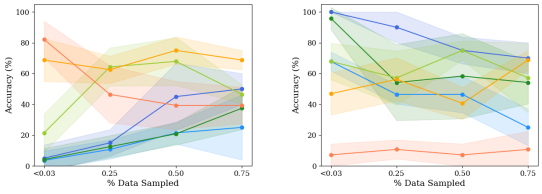


(a) y_1

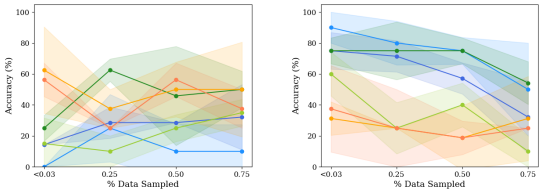


(b) y_2

Figure 43: QNLI dataset (Llama 3 8B)

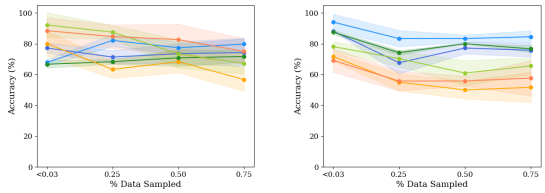


(a) y_1

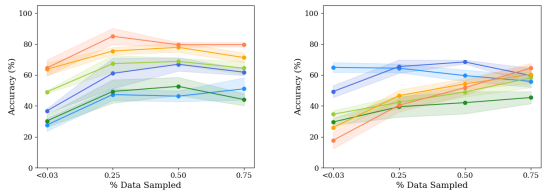


(b) y_2

Figure 44: WNLI dataset (Llama 3 8B)

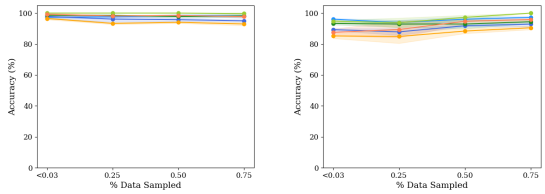


(a) y_1

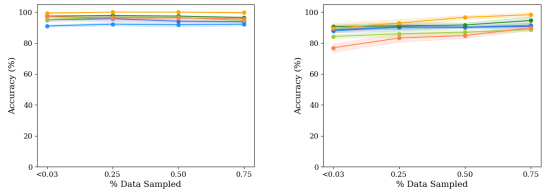


(b) y_2

Figure 45: MPRC dataset (Llama 3 8B)

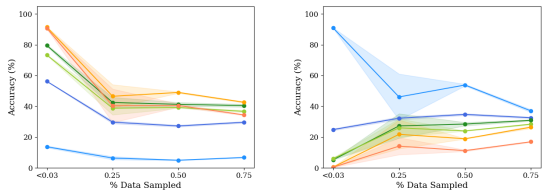


(a) y_1

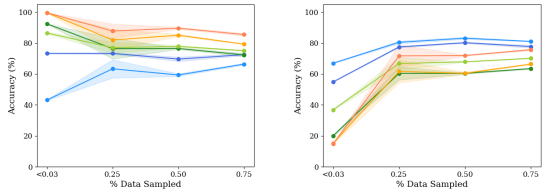


(b) y_2

Figure 46: SST-2 dataset (Llama 3 8B)

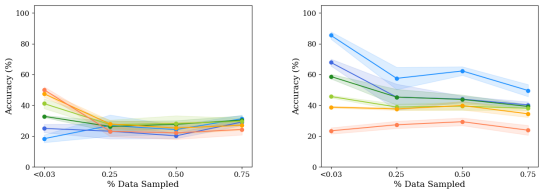


(a) y_1

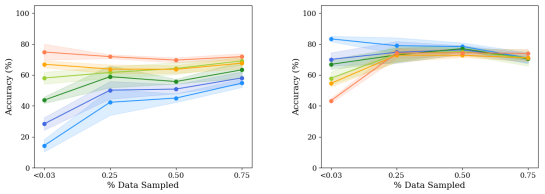


(b) y_2

Figure 47: Hans dataset (GPT Neo 2.7B)

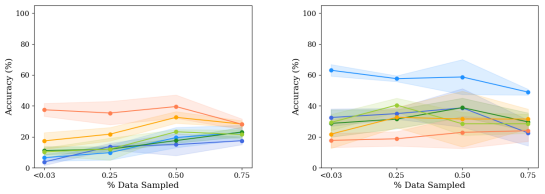


(a) y_1

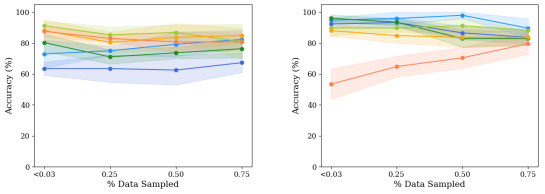


(b) y_2

Figure 48: PAWS-X_{EN} dataset (GPT Neo 2.7B)

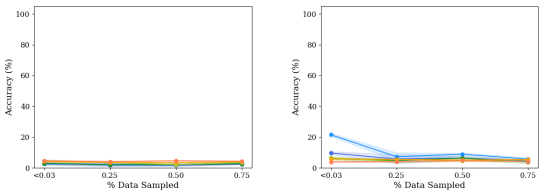


(a) y_1

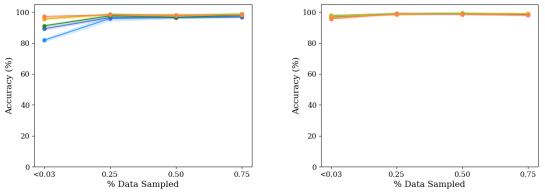


(b) y_2

Figure 49: RTE dataset (GPT Neo 2.7B)

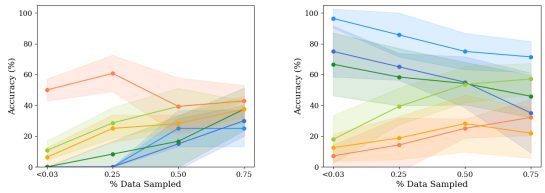


(a) y_1

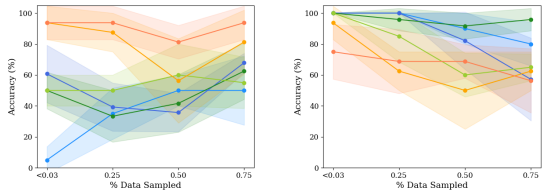


(b) y_2

Figure 50: QNLI dataset (GPT Neo 2.7B)



(a) y_1



(b) y_2

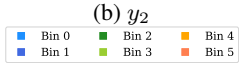
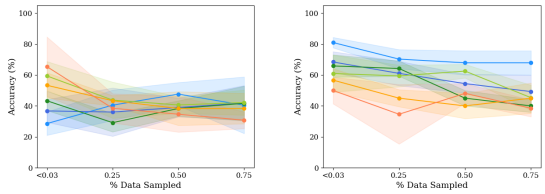
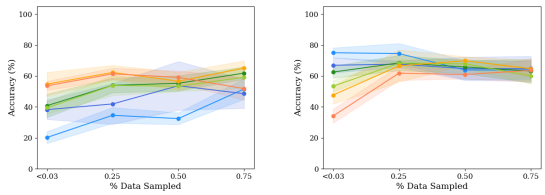


Figure 51: WNLI dataset (GPT Neo 2.7B)



(a) y_1



(b) y_2

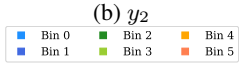
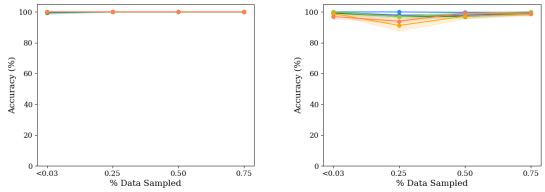
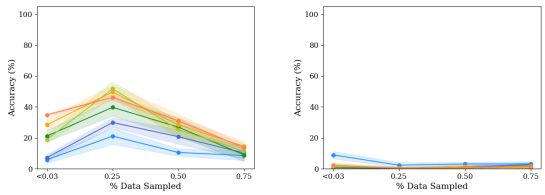


Figure 52: MPRC dataset (GPT Neo 2.7B)



(a) y_1



(b) y_2

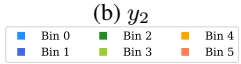
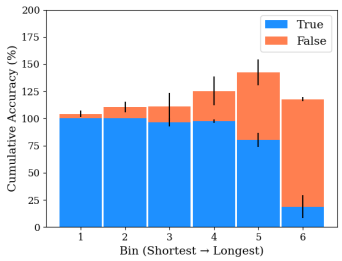


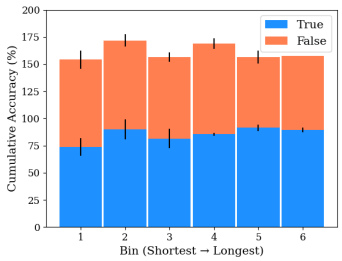
Figure 53: SST-2 dataset (GPT Neo 2.7B)

A.8 Additional Intervention Results

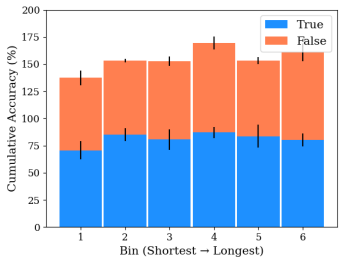
Each of the following figures shows validation set performance on a finetuned Llama 3 8B or GPT Neo 2.7B model exhibiting a length bias. For each figure, (a) shows finetuning performance prior to intervention). (b) and (c) show results on two debiasing conditions: ICL demonstrations ($k = 16$) sampled from the opposite lengths from what the model saw during finetuning (e.g y_1 long demonstrations, y_2 short demonstrations), and random sampling, respectively.



(a) Finetuning: y_1 (Blue) short demonstrations, y_2 (Orange) long demonstrations.

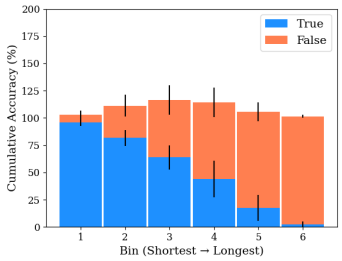


(b) Intervention: y_1 (Blue) long demonstrations, y_2 (Orange) short demonstrations.

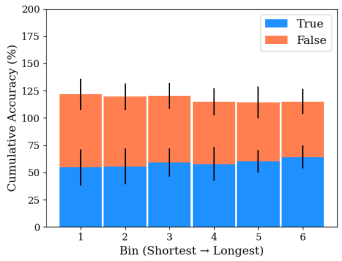


(c) Intervention: y_1 (Blue) and y_2 (Orange) demonstrations randomly sampled.

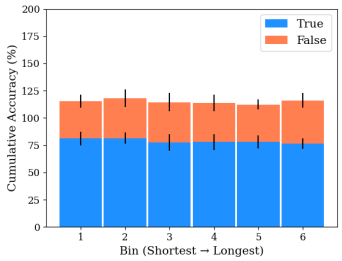
Figure 54: RTE (Llama 3 8B)



(a) Finetuning: y_1 (Blue) short demonstrations, y_2 (Orange) long demonstrations.

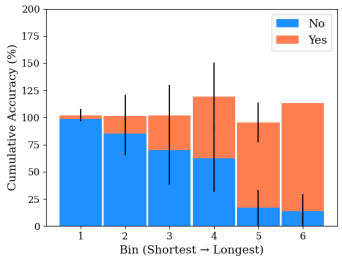


(b) Intervention: y_1 (Blue) long demonstrations, y_2 (Orange) short demonstrations.

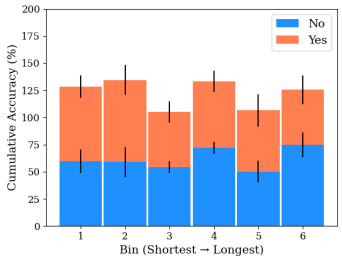


(c) Intervention: y_1 (Blue) and y_2 (Orange) demonstrations randomly sampled.

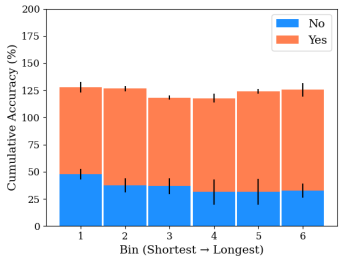
Figure 55: QNLI (Llama 3 8B)



(a) Finetuning: y_1 (Blue) short demonstrations, y_2 (Orange) long demonstrations.

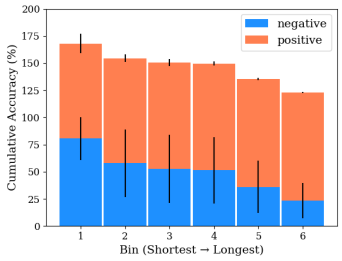


(b) Intervention: y_1 (Blue) long demonstrations, y_2 (Orange) short demonstrations.

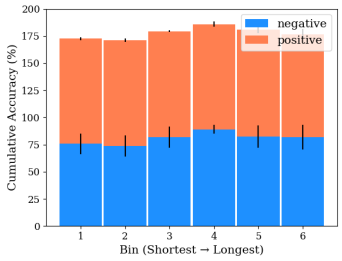


(c) Intervention: y_1 (Blue) and y_2 (Orange) demonstrations randomly sampled.

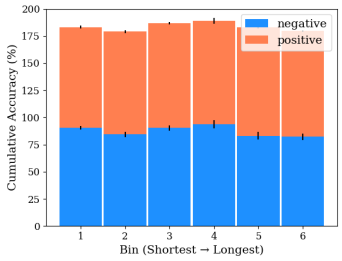
Figure 56: MRPC (Llama 3 8B)



(a) Finetuning: y_1 (Blue) short demonstrations, y_2 (Orange) long demonstrations.

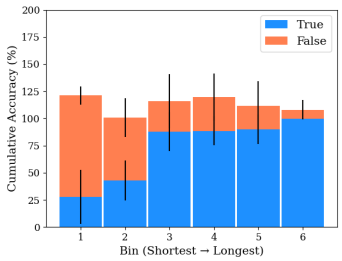


(b) Intervention: y_1 (Blue) long demonstrations, y_2 (Orange) short demonstrations.

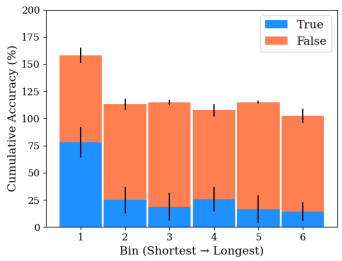


(c) Intervention: y_1 (Blue) and y_2 (Orange) demonstrations randomly sampled.

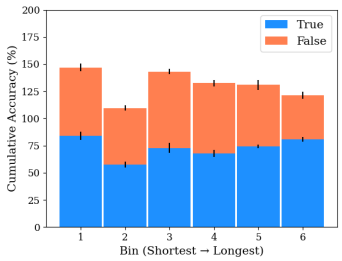
Figure 57: SST-2 (Llama 3 8B)



(a) Finetuning: y_1 (Blue) long demonstrations, y_2 (Orange) short demonstrations.

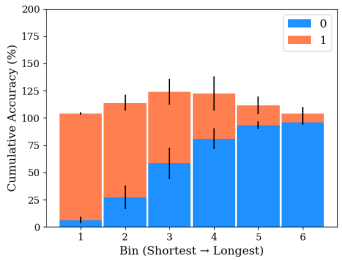


(b) Intervention: y_1 (Blue) short demonstrations, y_2 (Orange) long demonstrations.

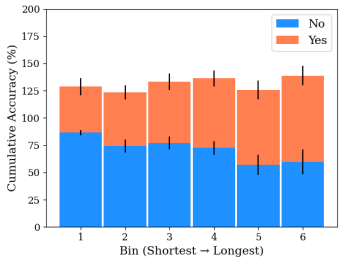


(c) Intervention: y_1 (Blue) and y_2 (Orange) demonstrations randomly sampled.

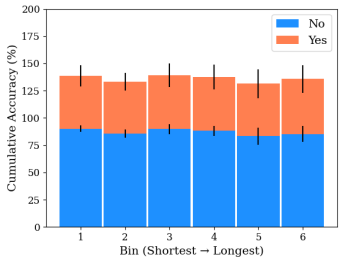
Figure 58: Hans (Llama 3 8B)



(a) Finetuning: y_1 (Blue) long demonstrations, y_2 (Orange) short demonstrations.

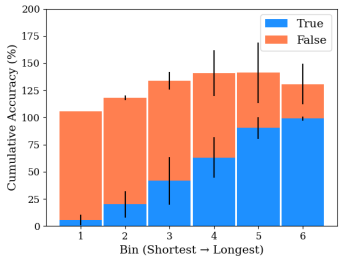


(b) Intervention: y_1 (Blue) short demonstrations, y_2 (Orange) long demonstrations.

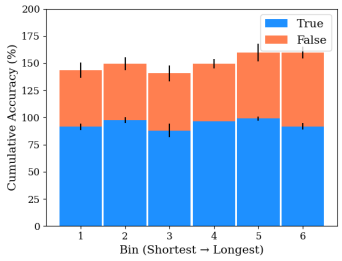


(c) Intervention: y_1 (Blue) and y_2 (Orange) demonstrations randomly sampled.

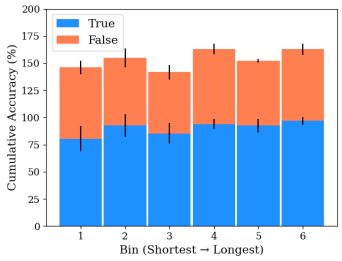
Figure 59: PAWS-X_{EN} (Llama 3 8B)



(a) Finetuning: y_1 (Blue) long demonstrations, y_2 (Orange) short demonstrations.

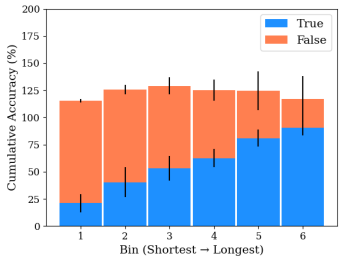


(b) Intervention: y_1 (Blue) short demonstrations, y_2 (Orange) long demonstrations.

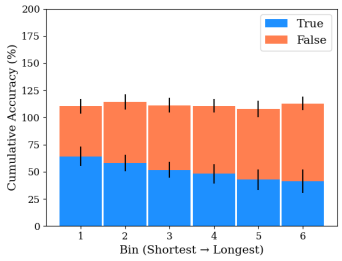


(c) Intervention: y_1 (Blue) and y_2 (Orange) demonstrations randomly sampled.

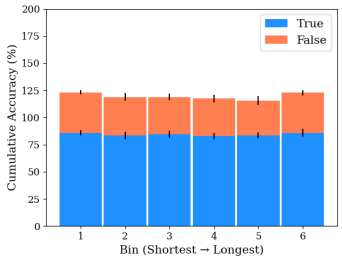
Figure 60: RTE (Llama 3 8B)



(a) Finetuning: y_1 (Blue) long demonstrations, y_2 (Orange) short demonstrations.

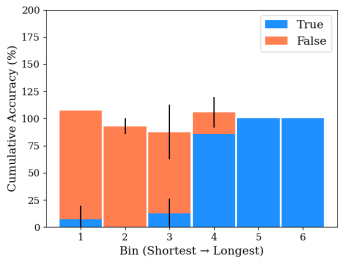


(b) Intervention: y_1 (Blue) short demonstrations, y_2 (Orange) long demonstrations.

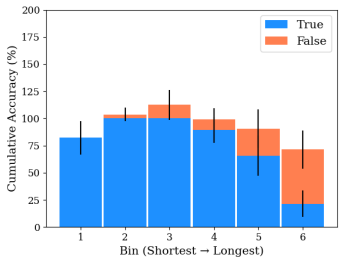


(c) Intervention: y_1 (Blue) and y_2 (Orange) demonstrations randomly sampled.

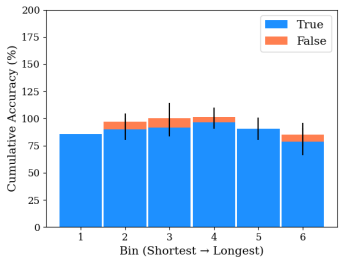
Figure 61: QNLI (Llama 3 8B)



(a) Finetuning: y_1 (Blue) long demonstrations, y_2 (Orange) short demonstrations.

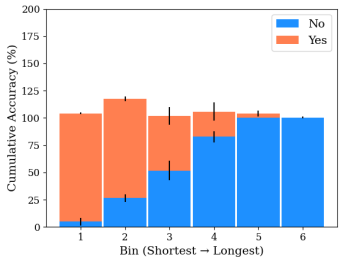


(b) Intervention: y_1 (Blue) short demonstrations, y_2 (Orange) long demonstrations.

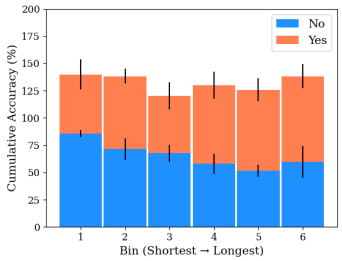


(c) Intervention: y_1 (Blue) and y_2 (Orange) demonstrations randomly sampled.

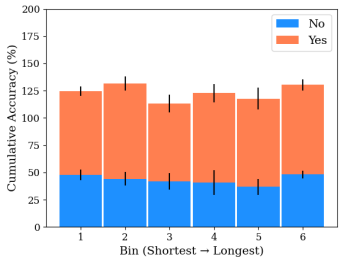
Figure 62: WNLI (Llama 3 8B)



(a) Finetuning: y_1 (Blue) long demonstrations, y_2 (Orange) short demonstrations.

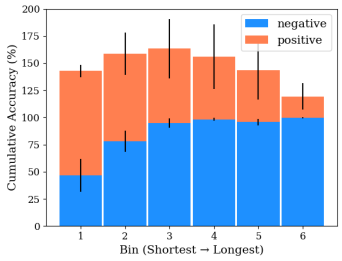


(b) Intervention: y_1 (Blue) short demonstrations, y_2 (Orange) long demonstrations.

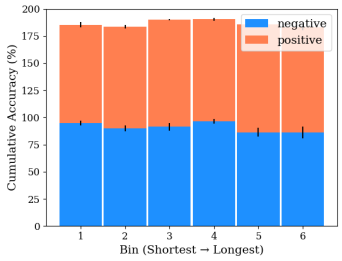


(c) Intervention: y_1 (Blue) and y_2 (Orange) demonstrations randomly sampled.

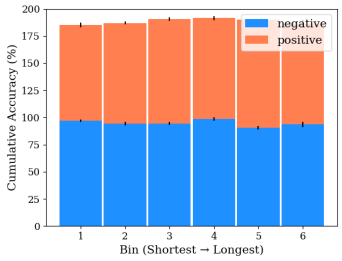
Figure 63: MRPC (Llama 3 8B)



(a) Finetuning: y_1 (Blue) long demonstrations, y_2 (Orange) short demonstrations.

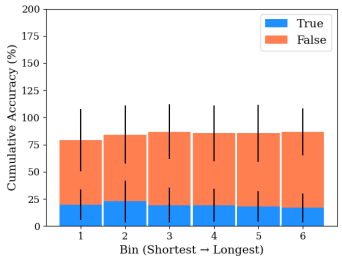


(b) Intervention: y_1 (Blue) short demonstrations, y_2 (Orange) long demonstrations.

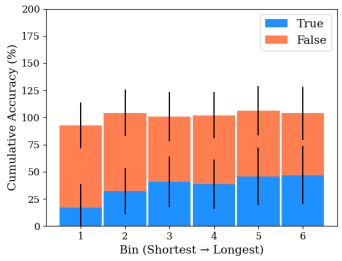


(c) Intervention: y_1 (Blue) and y_2 (Orange) demonstrations randomly sampled.

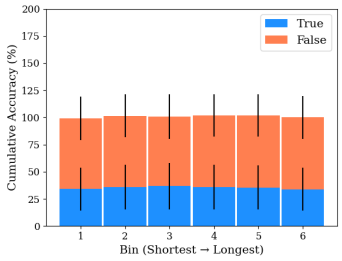
Figure 64: SST-2 (Llama 3 8B)



(a) Finetuning: y_1 (Blue) short demonstrations, y_2 (Orange) long demonstrations.

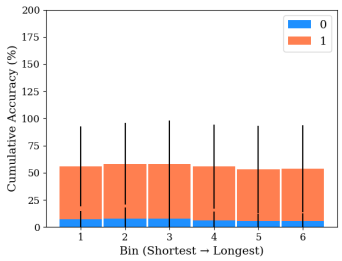


(b) Intervention: y_1 (Blue) long demonstrations, y_2 (Orange) short demonstrations.

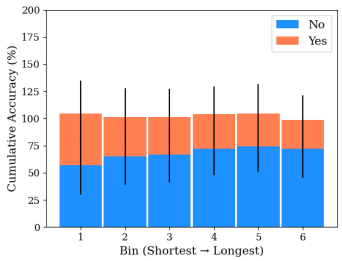


(c) Intervention: y_1 (Blue) and y_2 (Orange) demonstrations randomly sampled.

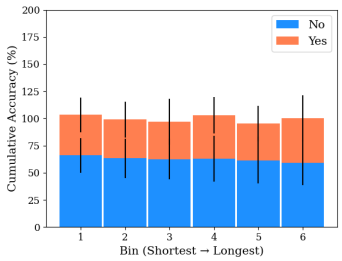
Figure 65: Hans (GPT Neo 2.7B)



(a) Finetuning: y_1 (Blue) short demonstrations, y_2 (Orange) long demonstrations.

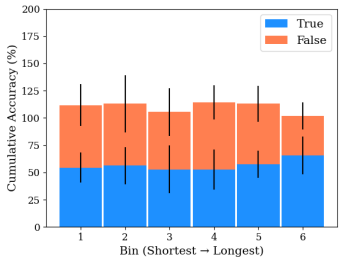


(b) Intervention: y_1 (Blue) long demonstrations, y_2 (Orange) short demonstrations.

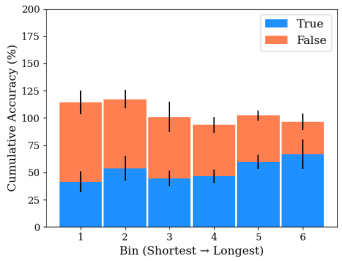


(c) Intervention: y_1 (Blue) and y_2 (Orange) demonstrations randomly sampled.

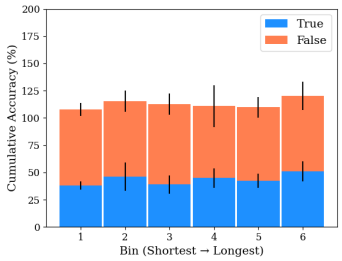
Figure 66: PAWS-X_{EN} (GPT Neo 2.7B)



(a) Finetuning: y_1 (Blue) short demonstrations, y_2 (Orange) long demonstrations.

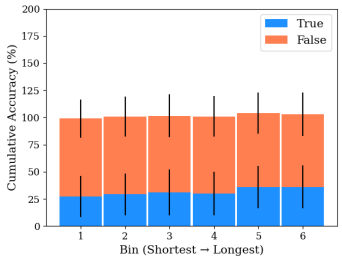


(b) Intervention: y_1 (Blue) long demonstrations, y_2 (Orange) short demonstrations.

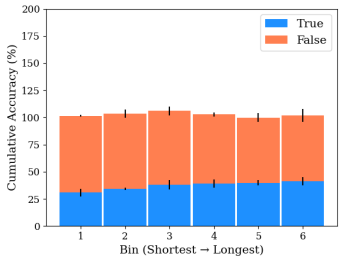


(c) Intervention: y_1 (Blue) and y_2 (Orange) demonstrations randomly sampled.

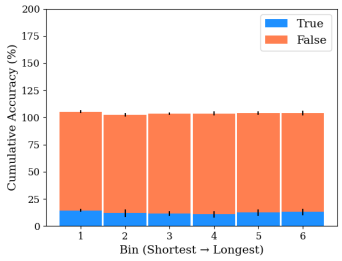
Figure 67: RTE (GPT Neo 2.7B)



(a) Finetuning: y_1 (Blue) short demonstrations, y_2 (Orange) long demonstrations.

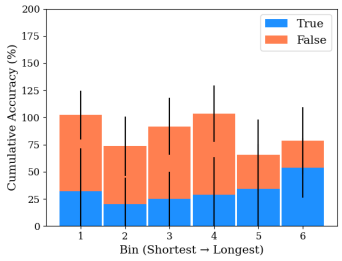


(b) Intervention: y_1 (Blue) long demonstrations, y_2 (Orange) short demonstrations.

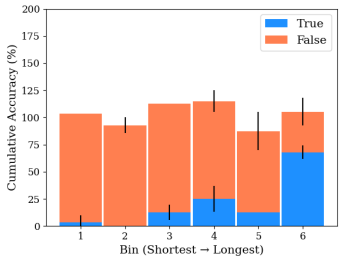


(c) Intervention: y_1 (Blue) and y_2 (Orange) demonstrations randomly sampled.

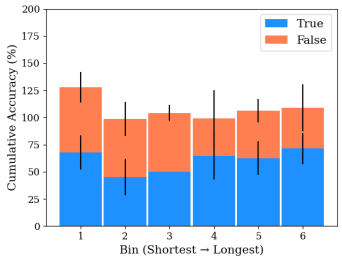
Figure 68: QNLI (GPT Neo 2.7B)



(a) Finetuning: y_1 (Blue) short demonstrations, y_2 (Orange) long demonstrations.

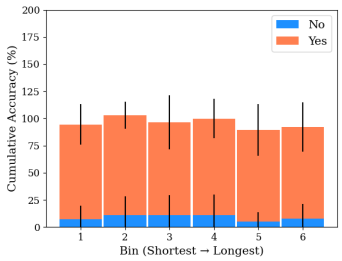


(b) Intervention: y_1 (Blue) long demonstrations, y_2 (Orange) short demonstrations.

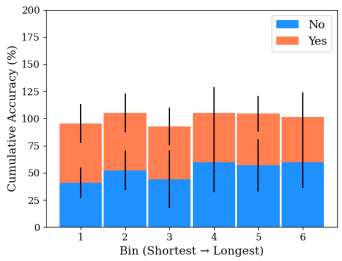


(c) Intervention: y_1 (Blue) and y_2 (Orange) demonstrations randomly sampled.

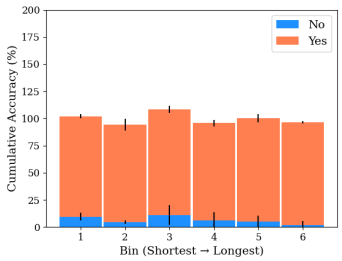
Figure 69: WNLI (GPT Neo 2.7B)



(a) Finetuning: y_1 (Blue) short demonstrations, y_2 (Orange) long demonstrations.

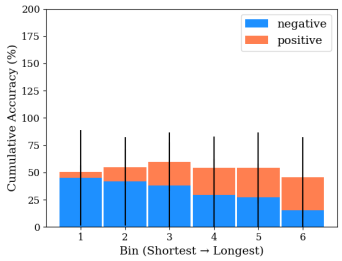


(b) Intervention: y_1 (Blue) long demonstrations, y_2 (Orange) short demonstrations.

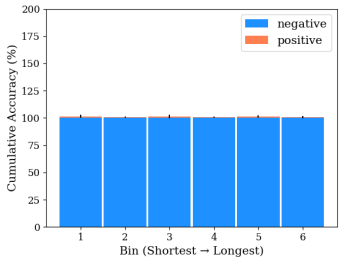


(c) Intervention: y_1 (Blue) and y_2 (Orange) demonstrations randomly sampled.

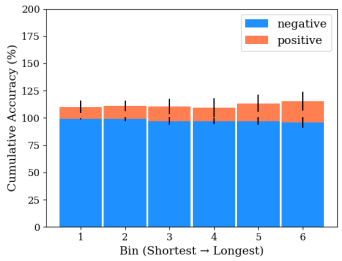
Figure 70: MRPC (GPT Neo 2.7B)



(a) Finetuning: y_1 (Blue) short demonstrations, y_2 (Orange) long demonstrations.

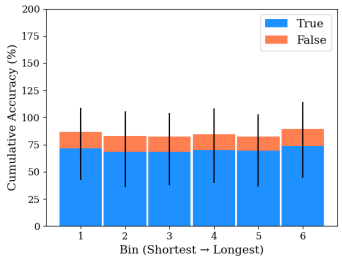


(b) Intervention: y_1 (Blue) long demonstrations, y_2 (Orange) short demonstrations.

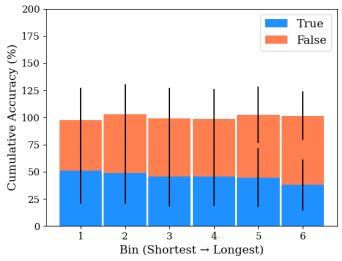


(c) Intervention: y_1 (Blue) and y_2 (Orange) demonstrations randomly sampled.

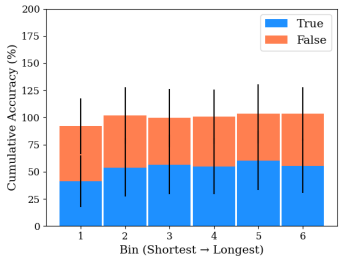
Figure 71: SST-2 (GPT Neo 2.7B)



(a) Finetuning: y_1 (Blue) long demonstrations, y_2 (Orange) short demonstrations.

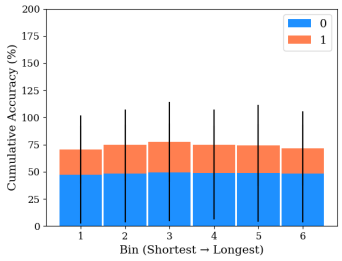


(b) Intervention: y_1 (Blue) short demonstrations, y_2 (Orange) long demonstrations.

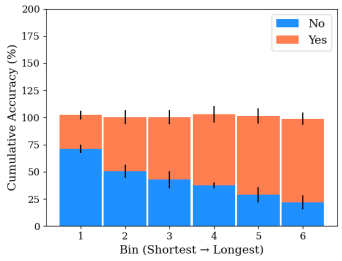


(c) Intervention: y_1 (Blue) and y_2 (Orange) demonstrations randomly sampled.

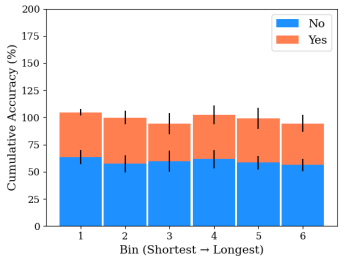
Figure 72: Hans (GPT Neo 2.7B)



(a) Finetuning: y_1 (Blue) long demonstrations, y_2 (Orange) short demonstrations.

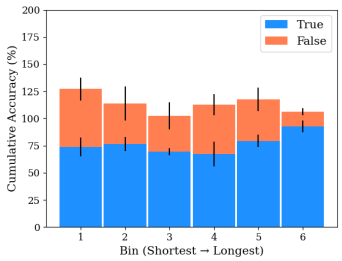


(b) Intervention: y_1 (Blue) short demonstrations, y_2 (Orange) long demonstrations.

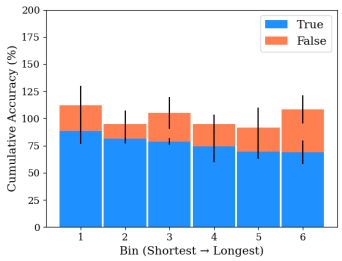


(c) Intervention: y_1 (Blue) and y_2 (Orange) demonstrations randomly sampled.

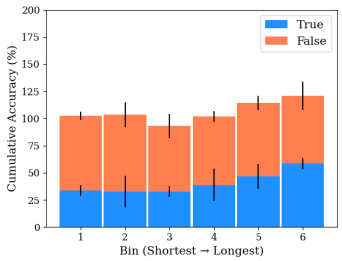
Figure 73: PAWS-X_{EN} (GPT Neo 2.7B)



(a) Finetuning: y_1 (Blue) long demonstrations, y_2 (Orange) short demonstrations.

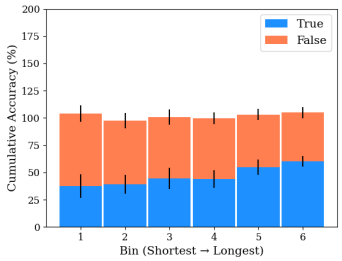


(b) Intervention: y_1 (Blue) short demonstrations, y_2 (Orange) long demonstrations.

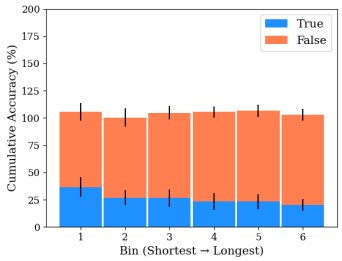


(c) Intervention: y_1 (Blue) and y_2 (Orange) demonstrations randomly sampled.

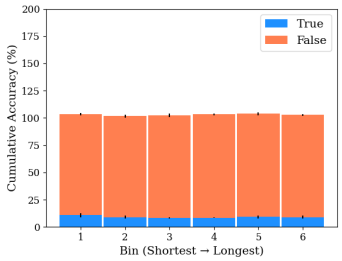
Figure 74: RTE (GPT Neo 2.7B)



(a) Finetuning: y_1 (Blue) long demonstrations, y_2 (Orange) short demonstrations.

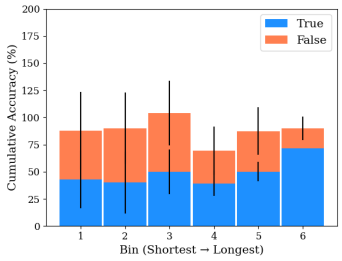


(b) Intervention: y_1 (Blue) short demonstrations, y_2 (Orange) long demonstrations.

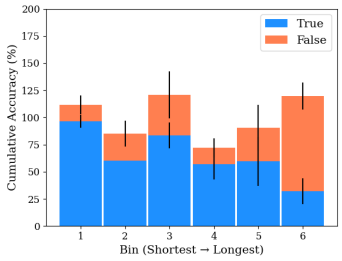


(c) Intervention: y_1 (Blue) and y_2 (Orange) demonstrations randomly sampled.

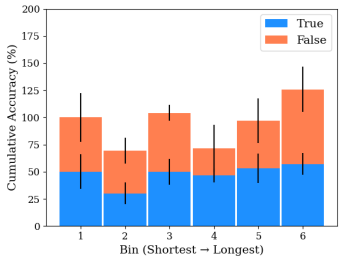
Figure 75: QNLI (GPT Neo 2.7B)



(a) Finetuning: y_1 (Blue) long demonstrations, y_2 (Orange) short demonstrations.

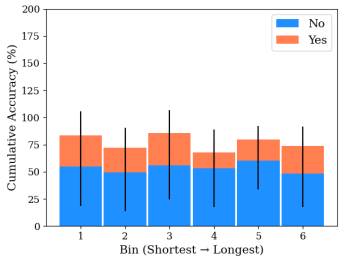


(b) Intervention: y_1 (Blue) short demonstrations, y_2 (Orange) long demonstrations.

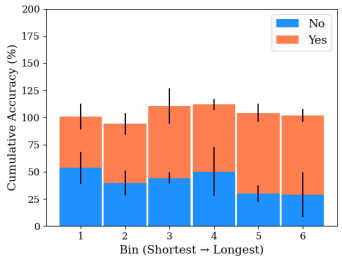


(c) Intervention: y_1 (Blue) and y_2 (Orange) demonstrations randomly sampled.

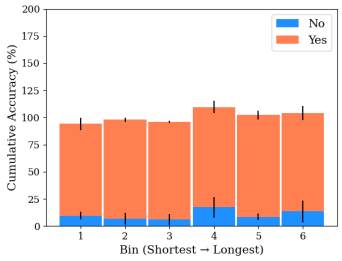
Figure 76: WNLI (GPT Neo 2.7B)



(a) Finetuning: y_1 (Blue) long demonstrations, y_2 (Orange) short demonstrations.

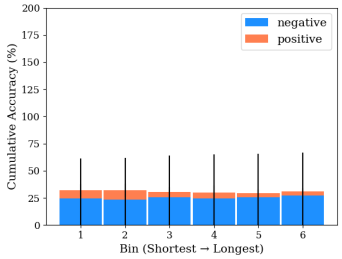


(b) Intervention: y_1 (Blue) short demonstrations, y_2 (Orange) long demonstrations.

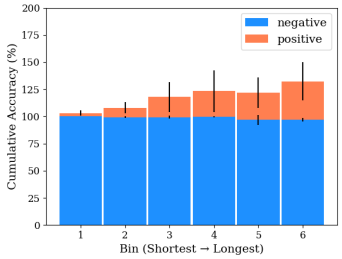


(c) Intervention: y_1 (Blue) and y_2 (Orange) demonstrations randomly sampled.

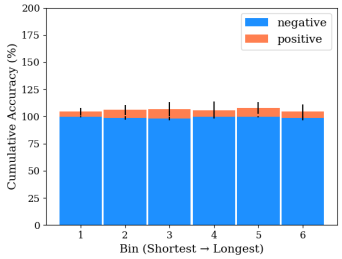
Figure 77: MRPC (GPT Neo 2.7B)



(a) Finetuning: y_1 (Blue) long demonstrations, y_2 (Orange) short demonstrations.



(b) Intervention: y_1 (Blue) short demonstrations, y_2 (Orange) long demonstrations.



(c) Intervention: y_1 (Blue) and y_2 (Orange) demonstrations randomly sampled.

Figure 78: SST-2 (GPT Neo 2.7B)